

Hierarchical DSTMs: Implementation and Inference

—♦—

In general, implementation and inference for DSTMs in the context of complicated hierarchical models is problem-specific. However, there are some common approaches that are used in both the EHM and BHM frameworks to find the filtering and forecast posterior distributions of the process and to perform inference on the parameters. We note that implementation approaches for DSTMs are rapidly evolving and still very much the subject of intensive research. Furthermore, these methodologies are linked very closely with the state of computing technology. In this chapter we give an overview, and we point to the many books, book chapters, and papers devoted to these implementation issues in the Bibliographic Notes in Section 8.5.

8.1 DSTM PROCESS: GENERAL IMPLEMENTATION AND INFERENCE

In this section, we (temporarily) assume that all parameters are fixed and *known*, and we focus our general discussion of implementation and inference on the state process. Thus, when denoting conditional distributions, we choose not to feature the parameters unless absolutely necessary.

Generally, the data model is

$$[\{\mathbf{Z}_t : t = 1, \dots, T\} | \{\mathbf{Y}_t : t = 0, \dots, T\}],$$

where $\mathbf{Z}_t$ is an m_t -dimensional vector of spatial observations at time t , and $\mathbf{Y}_t$ is an n -dimensional vector representation of the state process for n spatial locations at time t . The models discussed in Section 7.1 provide a rich source of examples.

At the next level, we specify the state-evolution distributions; in the case where the state process has first-order Markovian evolution, we specify

$$[\mathbf{Y}_0] \quad \text{and} \quad [\mathbf{Y}_t | \mathbf{Y}_{t-1}]; \quad t = 1, \dots, T.$$

Examples of such evolution distributions are given in Section 7.2. Typically, inference on the state process for such models is performed sequentially, and this will also be our focus here.

8.1.1 Sequential Implementation: General

Before discussing sequential approaches for state-process inference, we establish some useful notation. Let $\mathbf{Z}_{1:t} \equiv \{\mathbf{Z}_1, \dots, \mathbf{Z}_t\}$ and $\mathbf{Y}_{0:t} \equiv \{\mathbf{Y}_0, \dots, \mathbf{Y}_t\}$, for $t = 0, 1, \dots, T$. In addition, assume we employ a first-order Markov assumption, such that

$$[\mathbf{Y}_{0:T}] = [\mathbf{Y}_0] \prod_{t=1}^T [\mathbf{Y}_t | \mathbf{Y}_{t-1}], \quad (8.1)$$

where $[\mathbf{Y}_t | \mathbf{Y}_{t-1}]$ is the *evolution distribution*, and $[\mathbf{Y}_0]$ is an initial distribution that might in fact be based on all data up to and including $t = 0$, the reference starting time.

As discussed in Section 7.1, we assume that the observations in $\mathbf{Z}_{1:t}$ are conditionally independent, given $\mathbf{Y}_{0:t}$, and that for a fixed t the conditional-probability density functions of $\mathbf{Z}_t$, given $\mathbf{Y}_{0:t}$, in fact only depend on $\mathbf{Y}_t$; $t = 1, 2, \dots, T$. That is, the data model is

$$[\mathbf{Z}_{1:T} | \mathbf{Y}_{0:T}] = \prod_{t=1}^T [\mathbf{Z}_t | \mathbf{Y}_t], \quad (8.2)$$

where $[\mathbf{Z}_t | \mathbf{Y}_t]$ is sometimes called the measurement distribution.

Under the evolution-distribution assumptions (8.1) and the data-model assumptions (8.2), Bayes' rule can be used to find filtering and forecast distributions. That is, to predict $\mathbf{Y}_{0:T}$ in (8.1), given the observations $\mathbf{Z}_{1:T}$, we can write the posterior distribution of $\mathbf{Y}_{0:T}$ as

$$[\mathbf{Y}_{0:T} | \mathbf{Z}_{1:T}] \propto [\mathbf{Y}_0] \prod_{t=1}^T [\mathbf{Z}_t | \mathbf{Y}_t] [\mathbf{Y}_t | \mathbf{Y}_{t-1}]. \quad (8.3)$$

The key to the sequential approach is that, as new data become available, one seeks to update the previous filtering distribution. That is, the posterior distribution based on the data up to time $t - 1$ becomes the prior for the distribution for data up to time t . Specifically, we seek the *forecast distribution*,

$$[\mathbf{Y}_t | \mathbf{Z}_{1:t-1}], \quad (8.4)$$

since $\{\mathbf{Z}_{t+1}, \dots, \mathbf{Z}_T\}$ are independent of $\mathbf{Y}_t$ given $\mathbf{Y}_{t+1}$. Now, note that Bayes' rule gives,

$$[\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}] \propto [\mathbf{Y}_{t+1} | \mathbf{Y}_t, \mathbf{Z}_{1:t}] [\mathbf{Y}_t | \mathbf{Z}_{1:t}] = [\mathbf{Y}_{t+1} | \mathbf{Y}_t] [\mathbf{Y}_t | \mathbf{Z}_{1:t}], \quad (8.10)$$

where the first term on the right-hand side of (8.10) is just the evolution distribution, and the second term is the filtering distribution for time t . Thus, given the filtering and evolution distributions, a *sequential smoothing algorithm* written in pseudo code can be constructed as follows:

For $t = T - 1$ to 1

1. Obtain $[\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}]$ via (8.10), making use of the filtering distribution, $[\mathbf{Y}_t | \mathbf{Z}_{1:t}]$, and the evolution distribution, $[\mathbf{Y}_{t+1} | \mathbf{Y}_t]$.
2. Obtain the smoothing distribution, $[\mathbf{Y}_t | \mathbf{Z}_{1:T}]$, from (8.8), making use of the smoothing distribution from the subsequent time, $[\mathbf{Y}_{t+1} | \mathbf{Z}_{1:T}]$, obtained for the current time, t .

end

8.2 INFERENCE FOR THE DSTM PROCESS: LINEAR/GAUSSIAN MODELS

In general, it may be difficult to get explicit analytical formulations for the forecast, filtering, and smoothing distributions (8.6), (8.7), and (8.8), respectively. However, in the case of an observation operator that is linear, an evolution operator that is linear, and Gaussian error distributions, we can obtain the forecast, filtering, and smoothing distributions explicitly. This results in the well known *Kalman filter* and *Kalman smoother* recursions.

8.2.1 Kalman Filter

The *Kalman filter* is ideal for sequential updating, where the operators are linear and the error distributions are Gaussian (e.g., Kalman, 1960; Jazwinski, 1970; Anderson and Moore, 1979; West and Harrison, 1997; Wikle and Cressie, 1999). It can be derived from many different perspectives. Here, we use the general Bayesian formulas presented, for example, in Jazwinski (1970), Meinhold and Singpurwalla (1983), and West and Harrison (1997).

Define the conditional expectations for the filtering and forecast distributions as $\mathbf{Y}_{t|t} \equiv E[\mathbf{Y}_t | \mathbf{Z}_{1:t}]$ and $\mathbf{Y}_{t|t-1} \equiv E[\mathbf{Y}_t | \mathbf{Z}_{1:t-1}]$, respectively. Define the conditional error covariance matrices for filtering and forecasting, respectively, as

$$\mathbf{P}_{t|t} \equiv E[(\mathbf{Y}_t - \mathbf{Y}_{t|t})(\mathbf{Y}_t - \mathbf{Y}_{t|t})' | \mathbf{Z}_{1:t}]$$

and the *filtering distribution*,

$$[\mathbf{Y}_t | \mathbf{Z}_{1:t}], \quad (8.5)$$

for $t = 1, \dots, T$.

The *forecast distribution* (8.4) can be obtained by considering it in terms of the filtering distribution at the previous time:

$$\begin{aligned} [\mathbf{Y}_t | \mathbf{Z}_{1:t-1}] &= \int [\mathbf{Y}_t, \mathbf{Y}_{t-1} | \mathbf{Z}_{1:t-1}] d\mathbf{Y}_{t-1} \\ &= \int [\mathbf{Y}_t | \mathbf{Y}_{t-1}] [\mathbf{Y}_{t-1} | \mathbf{Z}_{1:t-1}] d\mathbf{Y}_{t-1}, \end{aligned} \quad (8.6)$$

where the second equality follows from the conditional-independence assumption. The first distribution in the integrand of (8.6) is the process-evolution distribution, and the second distribution is the filtering (posterior) distribution for the state at the previous time. Forecast distributions beyond one time step can be obtained similarly.

Now, the *filtering distribution* can be obtained by Bayes' rule:

$$\begin{aligned} [\mathbf{Y}_t | \mathbf{Z}_{1:t}] &= [\mathbf{Y}_t | \mathbf{Z}_t, \mathbf{Z}_{1:t-1}] \\ &\propto [\mathbf{Z}_t | \mathbf{Y}_t, \mathbf{Z}_{1:t-1}] [\mathbf{Y}_t | \mathbf{Z}_{1:t-1}] \\ &= [\mathbf{Z}_t | \mathbf{Y}_t] [\mathbf{Y}_t | \mathbf{Z}_{1:t-1}], \end{aligned} \quad (8.7)$$

where the first distribution on the right-hand side of (8.7) is obtained from the data model, and the second distribution is the forecast distribution (8.6). If we iterate between the filtering distribution and the forecast distribution as new data become available, we obtain the following sequences of distributions: $[\mathbf{Y}_1 | \mathbf{Z}_1], [\mathbf{Y}_2 | \mathbf{Z}_1], [\mathbf{Y}_2 | \mathbf{Z}_{1:2}], \dots, [\mathbf{Y}_T | \mathbf{Z}_{1:T-1}], [\mathbf{Y}_T | \mathbf{Z}_{1:T}]$. This sequential procedure is known as *filtering*.

The term *smoothing* refers to obtaining $[\mathbf{Y}_t | \mathbf{Z}_{1:T}]$, the distribution of the state at any time $t \in \{1, 2, \dots, T\}$, given all relevant data from time 1 to T, which includes data collected *after* time t up to time T . Obviously, for $t = T$, the smoothing distribution is just the filtering distribution.

We can write the smoothing distribution as

$$[\mathbf{Y}_t | \mathbf{Z}_{1:T}] = \int [\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:T}] [\mathbf{Y}_{t+1} | \mathbf{Z}_{1:T}] d\mathbf{Y}_{t+1}. \quad (8.8)$$

The first distribution on the right-hand side of (8.8) can be written as

$$[\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:T}] = [\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}], \quad (8.9)$$

and

$$\mathbf{P}_{t|t-1} \equiv E[(\mathbf{Y}_t - \mathbf{Y}_{t|t-1})(\mathbf{Y}_t - \mathbf{Y}_{t|t-1})' | \mathbf{Z}_{1:t-1}].$$

Consider the measurement distribution, $[\mathbf{Z}_t | \mathbf{Y}_t]$, given by the model,

$$\mathbf{Z}_t = \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t; \quad \boldsymbol{\varepsilon}_t \sim \text{ind. Gau}(\mathbf{0}, \mathbf{R}_t), \quad (8.11)$$

where $\mathbf{H}_t$ is the observation operator that maps the process to the observations, and $\mathbf{R}_t$ is the potentially time-varying measurement (or data) error covariance matrix, where the measurement errors are assumed to be independent across time.

Also, consider the evolution (or process) distribution, $[\mathbf{Y}_t | \mathbf{Y}_{t-1}]$, given by the VAR(1) model:

$$\mathbf{Y}_t = \mathbf{M}_t \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t; \quad \boldsymbol{\eta}_t \sim \text{ind. Gau}(\mathbf{0}, \mathbf{Q}_t), \quad (8.12)$$

where $\mathbf{M}_t$ is the (linear) model operator, or propagator matrix, that maps the evolution of the process in time, and $\mathbf{Q}_t$ is a covariance matrix representing temporally independent stochastic forcing, or features not resolved by the model. Typically, it is assumed that the measurement-error process and the evolution-error process are mutually independent for all time. We have also assumed that all error processes have mean zero. These assumptions of independence and zero means can be relaxed (e.g., Anderson and Moore, 1979).

Using standard conditional-expectation and conditional-variance calculations, the *forecast distribution* is found to be

$$\mathbf{Y}_t | \mathbf{Z}_{1:t-1} \sim \text{Gau}(\mathbf{Y}_{t|t-1}, \mathbf{P}_{t|t-1}),$$

with mean vector and covariance matrix given, respectively, by

$$\mathbf{Y}_{t|t-1} = E(E(\mathbf{Y}_t | \mathbf{Y}_{t-1}) | \mathbf{Z}_{1:t-1}) = E(\mathbf{M}_t \mathbf{Y}_{t-1} | \mathbf{Z}_{1:t-1}) = \mathbf{M}_t \mathbf{Y}_{t-1|t-1}, \quad (8.13)$$

and

$$\begin{aligned} \mathbf{P}_{t|t-1} &= E(\text{var}(\mathbf{Y}_t | \mathbf{Y}_{t-1}) | \mathbf{Z}_{1:t-1}) + \text{var}(E(\mathbf{Y}_t | \mathbf{Y}_{t-1}) | \mathbf{Z}_{1:t-1}) \\ &= E(\mathbf{Q}_t | \mathbf{Z}_{1:t-1}) + \text{var}(\mathbf{M}_t \mathbf{Y}_{t-1} | \mathbf{Z}_{1:t-1}) \\ &= \mathbf{Q}_t + \mathbf{M}_t \mathbf{P}_{t-1|t-1} \mathbf{M}_t'. \end{aligned} \quad (8.14)$$

In addition, the *filtering distribution* is given by

$$\mathbf{Y}_t | \mathbf{Z}_{1:t} \sim \text{Gau}(\mathbf{Y}_{t|t}, \mathbf{P}_{t|t}), \quad (8.15)$$

where

$$\mathbf{Y}_{t|t} = \mathbf{P}_{t|t}(\mathbf{H}_t' \mathbf{R}_t^{-1} \mathbf{Z}_t + \mathbf{P}_{t|t-1}^{-1} \mathbf{Y}_{t|t-1}), \quad (8.16)$$

and

$$\mathbf{P}_{t|t} = (\mathbf{H}_t' \mathbf{R}_t^{-1} \mathbf{H}_t + \mathbf{P}_{t|t-1}^{-1})^{-1}. \quad (8.17)$$

Using the Sherman–Morrison–Woodbury matrix identities (e.g., Henderson and Searle, 1981; Shumway and Stoffer, 2000, p. 317),

$$(\mathbf{A}' \mathbf{B}^{-1} \mathbf{A} + \mathbf{D}^{-1})^{-1} = \mathbf{D} - \mathbf{D} \mathbf{A}' (\mathbf{A} \mathbf{D} \mathbf{A}' + \mathbf{B})^{-1} \mathbf{A} \mathbf{D}, \quad (8.18)$$

$$(\mathbf{A}' \mathbf{B}^{-1} \mathbf{A} + \mathbf{D}^{-1})^{-1} \mathbf{A}' \mathbf{B}^{-1} = \mathbf{D} \mathbf{A}' (\mathbf{A} \mathbf{D} \mathbf{A}' + \mathbf{B})^{-1}, \quad (8.19)$$

we can write the mean and variance of (8.15) equivalently as

$$\mathbf{Y}_{t|t} = \mathbf{Y}_{t|t-1} + \mathbf{K}_t (\mathbf{Z}_t - \mathbf{H}_t \mathbf{Y}_{t|t-1}) \quad (8.20)$$

and

$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \mathbf{P}_{t|t-1}, \quad (8.21)$$

respectively, where $\mathbf{K}_t$ is called the *Kalman gain or gain matrix* and is given by

$$\mathbf{K}_t \equiv \mathbf{P}_{t|t-1} \mathbf{H}_t' (\mathbf{H}_t' \mathbf{P}_{t|t-1} \mathbf{H}_t + \mathbf{R}_t)^{-1}. \quad (8.22)$$

Assuming the parameter matrices, $\mathbf{H}_t$, $\mathbf{M}_t$, $\mathbf{Q}_t$, and $\mathbf{R}_t$ are known, for $t = 1, \dots, T$, and given initial conditions (or background state) $\mathbf{Y}_{0|0} \equiv \boldsymbol{\mu}_0$, $\mathbf{P}_{0|0} \equiv \boldsymbol{\Sigma}_0$, the following *Kalman filter algorithm* (written in pseudo code) can be used to obtain sequential estimates of the state, along with their mean squared prediction error matrices:

For $t = 1$ to T

1. Obtain the forecast-distribution mean, $\mathbf{Y}_{t|t-1}$, and covariance matrix, $\mathbf{P}_{t|t-1}$, from (8.13) and (8.14), respectively.
2. Obtain the gain, $\mathbf{K}_t$, and the filtering-distribution mean, $\mathbf{Y}_{t|t}$, and covariance matrix, $\mathbf{P}_{t|t}$, from (8.22), (8.20), and (8.21), respectively.

end

For time periods in which there are no observations, the filter-update step is simply skipped, and the filtering-distribution means and covariances are $\mathbf{Y}_{t|t} \equiv \mathbf{Y}_{t|t-1}$ and $\mathbf{P}_{t|t} \equiv \mathbf{P}_{t|t-1}$, respectively.

Example: As a simple example, assume that we have the univariate measurement model:

$$Z_t = Y_t + \epsilon_t, \quad \epsilon_t \sim iid \text{Gau}(0, 0.1), \quad (8.23)$$

where, in the notation introduced above, $R_t \equiv 0.1$ and $H_t \equiv 1$. Also assume we have the process model:

$$Y_t = 0.7Y_{t-1} + \eta_t, \quad \eta_t \sim iid \text{Gau}(0, 0.5), \quad (8.24)$$

where here $Q_t \equiv 0.5$ and $M_t \equiv 0.7$.

Given an initial condition, $Y_0 \sim \text{Gau}(0, 1)$, we simulated Y_t and Z_t for $t = 1, \dots, 100$. In addition, we deleted the data at times 40 to 43 and 80 to 83. The simulated process and data are shown in Figure 8.1. Our goal here is to make inference on the process (or “hidden state”) Y_t , using the filtering distribution. That is, we seek the distribution of Y_t given the data $\mathbf{Z}_{1:t}$; $t = 1, \dots, 100$. The Kalman filter is simple to compute for this univariate example. Figure 8.2 shows the estimate of the state process obtained by filtering, and Figure 8.3 shows the associated filtering-distribution variances, which are clearly larger during time periods where there are missing data.

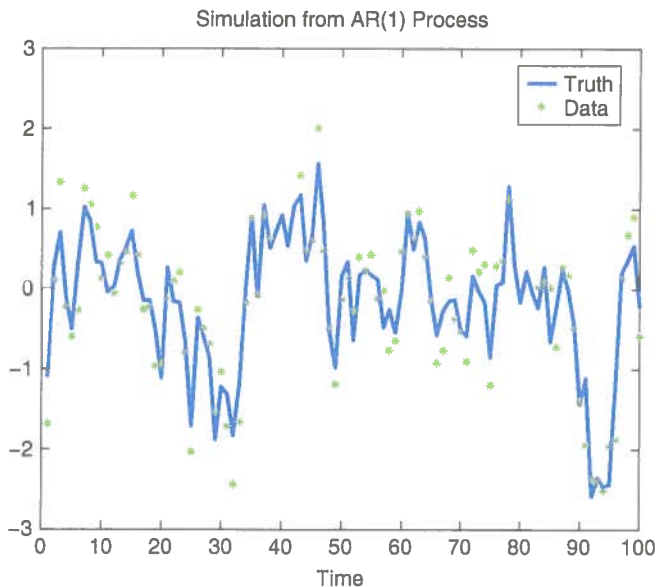


Figure 8.1 Simulated AR(1) process, or “truth” (solid blue line), and “data” (green dots).

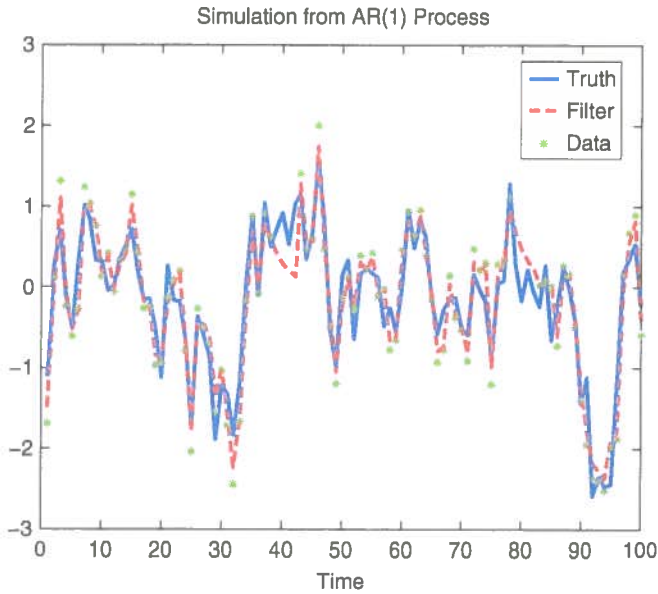


Figure 8.2 Kalman filter estimate of the state process (dashed red line), “truth” (solid blue line), and “data” (green dots).

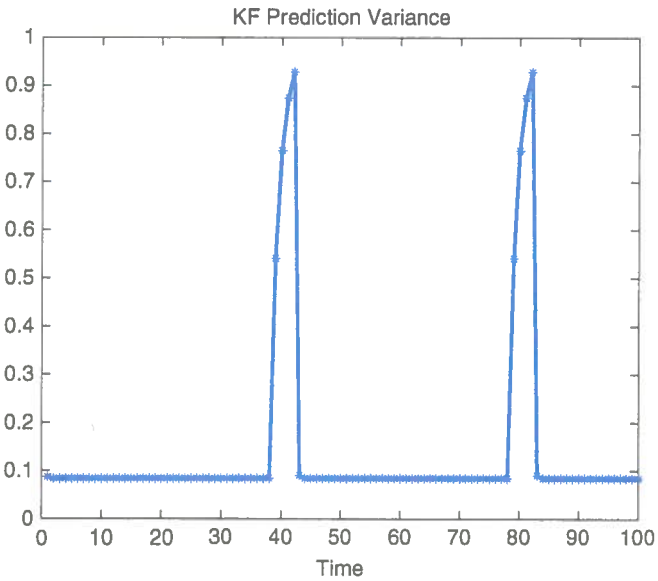


Figure 8.3 Kalman filter variance estimates associated with the simulated AR(1) process and data shown in Figure 8.2.

For more realistic examples featuring spatio-temporal data models and process models, see, for example, Wikle and Cressie (1999) and Kang, Cressie, and Shi (2010).

8.2.2 Kalman Smoother

As mentioned previously, in some situations there is interest in the distribution of $\mathbf{Y}_t$ at time $t \leq T$, given all the data up to T (i.e., both before and after time t), namely $[\mathbf{Y}_t | \mathbf{Z}_{1:T}]$; this is the *smoothing* (posterior) distribution of $\mathbf{Y}_t$. The smoothing distribution is useful for retrospective analyses (i.e., re-analyses) and is not something that one typically does “on-line.” As suggested by the general algorithm given in Section 8.1, we can use the Kalman filter and associated backwards-recursion formulas to obtain the smoothing distributions. This procedure, which is sometimes known as the *Kalman smoother*, can be derived from various perspectives. For extensive development, we refer the reader to one of the many excellent texts that discuss filtering and smoothing in detail, listed in the Bibliographic Notes in Section 8.5. We take an EHM approach in this subsection.

As with the Kalman-filter derivation, let $\mathbf{Y}_{t|T} \equiv E(\mathbf{Y}_t | \mathbf{Z}_{1:T})$ and $\mathbf{P}_{t|T} \equiv \text{var}(\mathbf{Y}_t | \mathbf{Z}_{1:T})$. We seek sequential formulas for $[\mathbf{Y}_t | \mathbf{Z}_{1:T}]$ and assume that we have access to the filtering distributions, $\mathbf{Y}_t | \mathbf{Z}_{1:t} \sim \text{Gau}(\mathbf{Y}_{t|t}, \mathbf{P}_{t|t})$, and the forecast distributions, $\mathbf{Y}_{t+1} | \mathbf{Z}_{1:t} \sim \text{Gau}(\mathbf{Y}_{t+1|t}, \mathbf{P}_{t+1|t})$, for $t = 1, \dots, T$. These are available from the Kalman-filter algorithm.

First, we can obtain $[\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}]$ from Bayes’ rule, as suggested by (8.10). This density is proportional to the product of two Gaussian densities:

$$[\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}] \propto \text{Gau}(\mathbf{Y}_{t|t}, \mathbf{P}_{t|t}) \text{Gau}(\mathbf{M}_{t+1} \mathbf{Y}_t, \mathbf{Q}_{t+1}), \quad (8.25)$$

where there is a slight abuse of notation by using $\text{Gau}(\cdot, \cdot)$ to represent Gaussian densities. By completing the square and using identities (8.18) and (8.19), this distribution is $\text{Gau}(\mathbf{u}_t, \mathbf{U}_t)$, where

$$\mathbf{u}_t \equiv \mathbf{Y}_{t|t} + \mathbf{J}_t (\mathbf{Y}_{t+1} - \mathbf{M}_{t+1} \mathbf{Y}_{t|t}), \quad (8.26)$$

$$\mathbf{U}_t \equiv \mathbf{P}_{t|t} - \mathbf{J}_t \mathbf{M}_{t+1} \mathbf{P}_{t|t}, \quad (8.27)$$

and

$$\mathbf{J}_t \equiv \mathbf{P}_{t|t} \mathbf{M}_{t+1}' (\mathbf{M}_{t+1} \mathbf{P}_{t|t} \mathbf{M}_{t+1}' + \mathbf{Q}_{t+1}^{-1}) = \mathbf{P}_{t|t} \mathbf{M}_{t+1}' \mathbf{P}_{t+1|t}^{-1}. \quad (8.28)$$

Next, we obtain the *smoothing distribution* from (8.8). Specifically,

$$\mathbf{Y}_t | \mathbf{Z}_{1:T} \sim \text{Gau}(\mathbf{Y}_{t|T}, \mathbf{P}_{t|T}), \quad (8.29)$$

where

$$\begin{aligned}\mathbf{Y}_{t|T} &= E(E(\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}) | \mathbf{Z}_{1:T}) \\ &= E(\mathbf{u}_t | \mathbf{Z}_{1:T}) = \mathbf{Y}_{t|t} + \mathbf{J}_t(\mathbf{Y}_{t+1|T} - \mathbf{Y}_{t+1|t})\end{aligned}\quad (8.30)$$

and

$$\begin{aligned}\mathbf{P}_{t|T} &= E(\text{var}(\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}) | \mathbf{Z}_{1:T}) + \text{var}(E(\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}) | \mathbf{Z}_{1:T}) \\ &= E(\mathbf{U}_t | \mathbf{Z}_{1:T}) + \text{var}(\mathbf{u}_t | \mathbf{Z}_{1:T}) \\ &= \mathbf{P}_{t|t} - \mathbf{J}_t \mathbf{M}_{t+1} \mathbf{P}_{t|t} + \mathbf{J}_t \text{var}(\mathbf{Y}_{t+1} | \mathbf{Z}_{1:T}) \mathbf{J}_t' \\ &= \mathbf{P}_{t|t} - \mathbf{J}_t \mathbf{P}_{t+1|t} \mathbf{J}_t' + \mathbf{J}_t \mathbf{P}_{t+1|T} \mathbf{J}_t' \\ &= \mathbf{P}_{t|t} + \mathbf{J}_t (\mathbf{P}_{t+1|T} - \mathbf{P}_{t+1|t}) \mathbf{J}_t'.\end{aligned}\quad (8.31)$$

Thus, given the forecast and filter estimates (actually, they are “predictions” rather than “estimates,” since we are inferring random quantities), $\mathbf{Y}_{t|t}$, $\mathbf{P}_{t|t}$, $\mathbf{Y}_{t+1|t}$, and $\mathbf{P}_{t+1|t}$, for $t = 0, \dots, T$, the *Kalman smoother algorithm* can be written in pseudo code as follows:

For $t = T - 1$ to 0

1. Obtain $\mathbf{J}_t$ from (8.28).
2. Obtain $\mathbf{Y}_{t|T}$ from (8.30).
3. Obtain $\mathbf{P}_{t|T}$ from (8.31).

end

Note that even when $T = 0$, the algorithm is still well defined because $\mathbf{Y}_{0|0} \equiv \boldsymbol{\mu}_0$ and $\mathbf{P}_{0|0} \equiv \boldsymbol{\Sigma}_0$. This smoothing algorithm is sometimes known as a *fixed-interval smoother*. Other smoothing algorithms (e.g., fixed-lag and fixed-point smoothers) can be implemented “on-line,” but they are not considered here (e.g., see Grewal and Andrews, 1993).

8.3 INFERENCE FOR THE DSTM PARAMETERS: LINEAR/GAUSSIAN MODELS

Recall that a critical assumption in the filtering and smoothing algorithms presented in the previous section is that the parameter matrices are known for all time t . Outside of rare engineering applications, this is almost never the case. It is fairly common that the measurement matrix $\mathbf{H}_t$ is known (and possibly the measurement-error covariance matrix, $\mathbf{R}_t$), but it is much less likely that the parameters associated with the evolution distribution are known. In the case where these parameter matrices are not time varying, one can, in principle,

estimate them. This is the empirical hierarchical modeling (EHM) framework discussed previously. One approach is to use method-of-moments estimates (e.g., Wikle and Cressie, 1999; Kang, Cressie, and Shi, 2010). Another approach is to maximize the likelihood numerically using Newton–Raphson (or similar) methods (e.g., Gupta and Mehra, 1974; Shumway and Stoffer, 2006, Section 6.3). Another likelihood-based approach is to use the Expectation-Maximization (EM) algorithm in this context (e.g., Shumway and Stoffer, 1982, 2006). Alternatively, the Bayesian hierarchical modeling (BHM) framework could be used, in which case the parameters are assumed to be random and are given prior distributions; then, estimation proceeds in a Bayesian fashion via the parameters’ posterior distributions. The following subsections consider the EM algorithm for the EHM approach, and a Gibbs sampler for the BHM approach.

8.3.1 EHM Implementation via the EM Algorithm

The basic EM algorithm is used to find maximum likelihood estimates in the presence of missing data (Dempster, Laird, and Rubin, 1977; McLachlan and Krishnan, 1997). In general, let the *complete data*, Z , be the union of the observations, Z_{obs} , and the missing data, Z_{mis} , so that $Z = (Z_{obs}, Z_{mis})$. If θ represents unknown parameters, then the complete data density is $[Z|\theta]$, and we write the complete likelihood as $L(\theta|Z) \equiv [Z|\theta]$; the loglikelihood is $\ln L(\theta|Z)$. Let $\hat{\theta}^{(i-1)}$ be the parameter estimate at the beginning of the i th iteration of the algorithm. Then, the algorithm consists of two steps:

- **E-Step:** Calculate $E(\ln L(\theta|Z)|Z_{obs}, \hat{\theta}^{(i-1)}) \equiv q(\theta|\hat{\theta}^{(i-1)})$.
- **M-Step:** Find θ that maximizes $q(\theta|\hat{\theta}^{(i-1)})$, and call this $\hat{\theta}^{(i)}$.

Thus, given a starting value, $\hat{\theta}^{(0)}$, iterating between the E-step and the M-step yields a sequence of estimates, $\{\hat{\theta}^{(i)} : i = 1, 2, \dots\}$. The algorithm is stopped when a given convergence criterion is met.

In the state-space setting given by (8.11) and (8.12), assume for illustration that the data-model matrix $\mathbf{H}_t$ is known and that the remainder of the parameters are *not time dependent*. Our exposition follows that of Shumway and Stoffer (2006, Section 6.3), with obvious notational differences. Let $\Theta \equiv \{\mu_0, \Sigma_0, \mathbf{Q}, \mathbf{R}, \mathbf{M}\}$ denote the set of parameters, where recall the initial condition, $\mathbf{Y}_{0|0} \sim \text{Gau}(\mu_0, \Sigma_0)$. In this context, the observed data are $\mathbf{Z}_{1:T}$, as before, and the “missing data” refer to the state process, $\mathbf{Y}_{0:T}$, which is, of course, unobservable by definition. The joint density of the complete data is then

$$[\mathbf{Z}_{1:T}, \mathbf{Y}_{0:T}|\Theta] = \left(\prod_{t=1}^T [\mathbf{Z}_t|\mathbf{Y}_t] \right) [\mathbf{Y}_0] \prod_{t=1}^T [\mathbf{Y}_t|\mathbf{Y}_{t-1}].$$

Thus, the complete-data, negative twice loglikelihood is (up to an additive constant)

$$\begin{aligned}
 -2 \ln L(\Theta | \mathbf{Z}_{1:T}, \mathbf{Y}_{0:T}) &= \ln |\Sigma_0| + (\mathbf{Y}_0 - \mu_0)' \Sigma_0^{-1} (\mathbf{Y}_0 - \mu_0) \\
 &+ T \ln |\mathbf{Q}| + \sum_{t=1}^T (\mathbf{Y}_t - \mathbf{M} \mathbf{Y}_{t-1})' \mathbf{Q}^{-1} (\mathbf{Y}_t - \mathbf{M} \mathbf{Y}_{t-1}) \\
 &+ T \ln |\mathbf{R}| + \sum_{t=1}^T (\mathbf{Z}_t - \mathbf{H}_t \mathbf{Y}_t)' \mathbf{R}^{-1} (\mathbf{Z}_t - \mathbf{H}_t \mathbf{Y}_t).
 \end{aligned} \tag{8.32}$$

Now, the E-step of the EM algorithm requires the calculation of the expectation,

$$\begin{aligned}
 q(\Theta | \hat{\Theta}^{(i-1)}) &\equiv E(-2 \ln L(\Theta | \mathbf{Z}_{1:T}, \mathbf{Y}_{1:T}) | \mathbf{Z}_{1:T}, \hat{\Theta}^{(i-1)}) \\
 &= \ln |\Sigma_0| + \text{tr} \left\{ \Sigma_0^{-1} [\mathbf{P}_{0|T} + (\mathbf{Y}_{0|T} - \mu_0)(\mathbf{Y}_{0|T} - \mu_0)'] \right\} \\
 &+ T \ln |\mathbf{Q}| + \text{tr} \left\{ \mathbf{Q}^{-1} [\mathbf{S}_{11} - \mathbf{S}_{10} \mathbf{M}' - \mathbf{M} \mathbf{S}'_{10} + \mathbf{M} \mathbf{S}_{00} \mathbf{M}'] \right\} + T \ln |\mathbf{R}| \\
 &+ \text{tr} \left\{ \mathbf{R}^{-1} \sum_{t=1}^T [(\mathbf{Z}_t - \mathbf{H}_t \mathbf{Y}_{t|T})(\mathbf{Z}_t - \mathbf{H}_t \mathbf{Y}_{t|T})' + \mathbf{H}_t \mathbf{P}_{t|T} \mathbf{H}_t'] \right\},
 \end{aligned} \tag{8.33}$$

for

$$\begin{aligned}
 \mathbf{S}_{11} &\equiv \sum_{t=1}^T (\mathbf{P}_{t|T} + \mathbf{Y}_{t|T} \mathbf{Y}_{t|T}') \\
 \mathbf{S}_{00} &\equiv \sum_{t=1}^T (\mathbf{P}_{t-1|T} + \mathbf{Y}_{t-1|T} \mathbf{Y}_{t-1|T}') \\
 \mathbf{S}_{10} &\equiv \sum_{t=1}^T (\mathbf{P}_{t,t-1|T} + \mathbf{Y}_{t|T} \mathbf{Y}_{t-1|T}'),
 \end{aligned}$$

where, in general, $\mathbf{P}_{t_1, t_2|T} \equiv E((\mathbf{Y}_{t_1} - \mathbf{Y}_{t_1|T})(\mathbf{Y}_{t_2} - \mathbf{Y}_{t_2|T})' | \mathbf{Z}_{1:T})$ and $\mathbf{P}_{t|T} \equiv \mathbf{P}_{t,t|T}$. The right-hand side of (8.33) is dependent on $\hat{\Theta}^{(i-1)}$, although for economy of notation this is *not* made explicit. For example, $\mathbf{Y}_{t|T} = E(\mathbf{Y}_t | \mathbf{Z}_{1:T})$ depends on Θ ; when it appears on the right-hand side of (8.33), it should be understood that it is evaluated at $\Theta = \hat{\Theta}^{(i-1)}$.

The M-step then proceeds by minimizing $q(\Theta|\hat{\Theta}^{(i-1)})$, as given by (8.33), with respect to the parameters Θ . This gives

$$\hat{\mathbf{M}}^{(i)} = \mathbf{S}_{10}\mathbf{S}_{00}^{-1} \quad (8.34)$$

$$\hat{\mathbf{Q}}^{(i)} = (1/T)(\mathbf{S}_{11} - \mathbf{S}_{10}\mathbf{S}_{00}^{-1}\mathbf{S}_{10}') \quad (8.35)$$

$$\hat{\mathbf{R}}^{(i)} = (1/T) \sum_{t=1}^T ((\mathbf{Z}_t - \mathbf{H}_t\mathbf{Y}_{t|T})(\mathbf{Z}_t - \mathbf{H}_t\mathbf{Y}_{t|T})' + \mathbf{H}_t\mathbf{P}_{t|T}\mathbf{H}_t'), \quad (8.36)$$

and

$$\hat{\boldsymbol{\mu}}_0^{(i)} = \mathbf{Y}_{0|T}. \quad (8.37)$$

Although we follow the tradition of estimating $\boldsymbol{\mu}_0$ and fixing $\boldsymbol{\Sigma}_0$ (e.g., Shumway and Stoffer, 2000, p.325), we note that one could update $\boldsymbol{\Sigma}_0$ by setting it equal to $\mathbf{P}_{0|T}$ (e.g., Shumway and Stoffer, 2006, p. 344). Thus, in our case, $\hat{\Theta}^{(i)} \equiv \{\hat{\boldsymbol{\mu}}_0^{(i)}, \hat{\mathbf{Q}}^{(i)}, \hat{\mathbf{R}}^{(i)}, \hat{\mathbf{M}}^{(i)}\}$, and $\boldsymbol{\Sigma}_0$ is fixed. However, before $\hat{\Theta}^{(i)}$ can be calculated, we still have to evaluate $\mathbf{P}_{t,t-1|T}$ in $\mathbf{S}_{10}$, which can be accomplished by the *lag-one covariance smoother* (Shumway and Stoffer, 2006, Property P6.3). Given $\mathbf{K}_t$ defined by (8.22), $\mathbf{J}_t$ defined by (8.28), and $\mathbf{P}_{t|t}$ defined by (8.21) evaluated at $\hat{\Theta}^{(i-1)}$, for $t = 0, \dots, T$, we first calculate $\mathbf{P}_{T,T-1|T} = (\mathbf{I} - \mathbf{K}_T\mathbf{H}_T)\mathbf{M}\mathbf{P}_{T-1|T-1}$. Then, for $t = T, T-1, \dots, 2$,

$$\mathbf{P}_{t-1,t-2|T} = \mathbf{P}_{t-1|t-1}\mathbf{J}_{t-2}' + \mathbf{J}_{t-1}(\mathbf{P}_{t,t-1|T} - \mathbf{M}\mathbf{P}_{t-1|t-1})\mathbf{J}_{t-2}'.$$

Then the *EM algorithm* can be written in pseudo code as follows:

Select starting values, $\hat{\Theta}^{(0)} = \{\hat{\boldsymbol{\mu}}_0^{(0)}, \hat{\mathbf{Q}}^{(0)}, \hat{\mathbf{R}}^{(0)}, \hat{\mathbf{M}}^{(0)}\}$, and choose $\boldsymbol{\Sigma}_0$.

For $i = 1, 2, \dots$ (until convergence)

1. **E-step:** Use $\hat{\Theta}^{(i-1)}$ in the Kalman filter, Kalman smoother, and lag-one covariance smoother to obtain $\mathbf{Y}_{t|T}$, $\mathbf{P}_{t|T}$, $\mathbf{P}_{t,t-1|T}$, for $t = 1, \dots, T$. Use these quantities to calculate $\mathbf{S}_{00}$, $\mathbf{S}_{11}$, and $\mathbf{S}_{10}$.
2. **M-step:** Update estimates $\hat{\Theta}^{(i)}$, specifically, $\hat{\boldsymbol{\mu}}_0^{(i)}$, $\hat{\mathbf{Q}}^{(i)}$, $\hat{\mathbf{R}}^{(i)}$, and $\hat{\mathbf{M}}^{(i)}$, using (8.37), (8.35), (8.36), and (8.34), respectively.

end

We also note that sometimes the M-step update is difficult to obtain; this occurs, for example, in the case of highly parameterized propagator and covariance matrices in state-space models. In this situation, a general EM (GEM) approach can be taken, and the M-step can be replaced with several simple conditional maximization steps (McLachlan and Krishnan, 1997). Alternatively, for

scalar parameters, it may be possible to use a Newton–Raphson step within the GEM (McLachlan and Krishnan, 1997). Xu and Wikle (2007) discuss the application of these approaches to structured matrix parameters in DSTMs.

Although we do not explicitly describe algorithms to obtain standard errors for the EHM parameter estimates, they can be computed in various ways. For example, it may be possible to evaluate the Hessian matrix after convergence (Shumway and Stoffer, 2006). Alternatively, it may be possible to obtain the standard errors by perturbing the likelihood function and using numerical differentiation (e.g., Shumway and Stoffer, 2006; Tanner, 1996). However, although the likelihood-based parameter estimates are consistent and asymptotically normal, the asymptotic results may not be applicable in spatio-temporal applications where the sample sizes are relatively small. In that case, a bootstrap procedure may be appropriate, and it is straightforward to implement. For example, Stoffer and Wall (1991) describe a simple bootstrap-sampling algorithm, for parameter estimates in general state-space models, that is appropriate for the spatio-temporal setting discussed here. In addition, Wall and Stoffer (2002) describe how bootstrap resampling in this context can also give appropriate estimates of conditional forecast accuracy.

8.3.2 BHM Implementation via the Gibbs Sampler

Assume the data model:

$$\mathbf{Z}_t = \mathbf{H}_t \mathbf{Y}_t + \boldsymbol{\varepsilon}_t; \quad \boldsymbol{\varepsilon}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{R}), \quad (8.38)$$

for $t = 1, \dots, T$, and the process model:

$$\mathbf{Y}_t = \mathbf{M} \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t; \quad \boldsymbol{\eta}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}), \quad (8.39)$$

$$\mathbf{Y}_0 \sim \text{Gau}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0). \quad (8.40)$$

Now, since we are considering inference using a BHM, we must specify prior distributions for parameters. For example, assume the parameter model,

$$\mathbf{R}^{-1} \sim W((\nu_R \mathbf{C}_R)^{-1}, \nu_R) \quad (8.41)$$

$$\mathbf{Q}^{-1} \sim W((\nu_Q \mathbf{C}_Q)^{-1}, \nu_Q) \quad (8.42)$$

$$\mathbf{m} \equiv \text{vec}(\mathbf{M}) \sim \text{Gau}(\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m), \quad (8.43)$$

where $W(\cdot, \cdot)$ denotes a Wishart distribution. For now, we assume that the (hyper)parameters $\boldsymbol{\mu}_0$, $\boldsymbol{\Sigma}_0$, ν_R , $\mathbf{C}_R$, ν_Q , $\mathbf{C}_Q$, $\boldsymbol{\mu}_m$, and $\boldsymbol{\Sigma}_m$ are fixed and “known.” One advantage of the hierarchical approach is that we could assign prior distributions to these parameters as well. While (8.41), (8.42), and (8.43) are chosen for computational ease, other prior distributions could be chosen (e.g., see West and Harrison, 1997; Section 7.5).

We seek the posterior distribution,

$$\begin{aligned}
 & [Y_0, Y_1, \dots, Y_T, \mathbf{M}, \mathbf{Q}, \mathbf{R} | Z_{1:T}] \\
 & \propto \left(\prod_{t=1}^T [Z_t | Y_t, \mathbf{R}] \right) [Y_0] \left(\prod_{t=1}^T [Y_t | Y_{t-1}, \mathbf{M}, \mathbf{Q}] \right) [\mathbf{R}][\mathbf{Q}][\mathbf{M}], \quad (8.44)
 \end{aligned}$$

where, for convenience, we have assumed that the parameter model is based on independent distributions. Unfortunately, the normalizing constant for (8.44) cannot be found analytically. However, Markov chain Monte Carlo (MCMC) samples from this posterior distribution can be obtained from a Gibbs sampler. This requires the following full conditional distributions:

$$\begin{aligned}
 [Y_0 | \cdot] & \propto [Y_1 | Y_0, \mathbf{M}, \mathbf{Q}] [Y_0], \\
 [Y_t | \cdot] & \propto [Z_t | Y_t, \mathbf{R}] [Y_{t+1} | Y_t, \mathbf{M}, \mathbf{Q}] [Y_t | Y_{t-1}, \mathbf{M}, \mathbf{Q}], \quad t = 1, \dots, T-1, \\
 [Y_T | \cdot] & \propto [Z_T | Y_T, \mathbf{R}] [Y_T | Y_{T-1}, \mathbf{M}, \mathbf{Q}], \\
 [\mathbf{R} | \cdot] & \propto \prod_{t=1}^T [Z_t | Y_t, \mathbf{R}] [\mathbf{R}], \\
 [\mathbf{Q} | \cdot] & \propto \prod_{t=1}^T [Y_t | Y_{t-1}, \mathbf{M}, \mathbf{Q}] [\mathbf{Q}], \\
 [\mathbf{M} | \cdot] & \propto \prod_{t=1}^T [Y_t | Y_{t-1}, \mathbf{M}, \mathbf{Q}] [\mathbf{M}],
 \end{aligned}$$

where “ $\cdot$ ” in the conditioning set denotes all quantities on the left-hand side of (8.44), except the variable whose conditional distribution we are considering.

All of the full conditional distributions implied by (8.38)–(8.43) are “conjugate,” and this greatly simplifies the derivation of these distributions. To illustrate what this means, consider the case where we have an n_w -dimensional Gaussian distribution, $\mathbf{w} | \mathbf{x} \sim \text{Gau}(\mathbf{A}\mathbf{x}, \Sigma_w)$, and an n_x -dimensional Gaussian distribution, $\mathbf{x} \sim \text{Gau}(\mathbf{B}\boldsymbol{\mu}, \Sigma_x)$; we are interested in the conditional distribution, $[\mathbf{x} | \mathbf{w}] \propto [\mathbf{w} | \mathbf{x}][\mathbf{x}]$. First,

$$\begin{aligned}
 [\mathbf{x} | \mathbf{w}] & \propto \exp\{-(1/2)[(\mathbf{w} - \mathbf{A}\mathbf{x})' \Sigma_w^{-1} (\mathbf{w} - \mathbf{A}\mathbf{x}) + (\mathbf{x} - \mathbf{B}\boldsymbol{\mu})' \Sigma_x^{-1} (\mathbf{x} - \mathbf{B}\boldsymbol{\mu})]\} \\
 & \propto \exp\{-(1/2)[\mathbf{x}(\mathbf{A}' \Sigma_w^{-1} \mathbf{A} + \Sigma_x^{-1})\mathbf{x}' - 2(\mathbf{w}' \Sigma_w^{-1} \mathbf{A} + \boldsymbol{\mu}' \mathbf{B}' \Sigma_x^{-1})\mathbf{x}]\},
 \end{aligned}$$

where it is understood that here $\mathbf{x} \in \mathbb{R}^{n_x}$ and $\mathbf{w} \in \mathbb{R}^{n_w}$. Now, using the fact that $[\mathbf{v}] \propto \exp\{-1/2(\mathbf{v}' \mathbf{D} \mathbf{v} - 2\mathbf{b}' \mathbf{v})\}$ implies that $\mathbf{v} \sim \text{Gau}(\mathbf{D}^{-1} \mathbf{b}, \mathbf{D}^{-1})$, which

is obtained by completing the square, the general Gaussian/Gaussian result for the conditional distribution follows:

$$\mathbf{x}|\mathbf{w} \sim \text{Gau}\left((\mathbf{A}'\Sigma_w^{-1}\mathbf{A} + \Sigma_x^{-1})^{-1}(\mathbf{w}'\Sigma_w^{-1}\mathbf{A} + \boldsymbol{\mu}'\mathbf{B}'\Sigma_x^{-1})', (\mathbf{A}'\Sigma_w^{-1}\mathbf{A} + \Sigma_x^{-1})^{-1}\right). \quad (8.45)$$

Similarly, if $\mathbf{x}_i|\Sigma_x \sim \text{Gau}(\boldsymbol{\mu}, \Sigma_x)$, for $i = 1, \dots, T$, and $\Sigma_x^{-1} \sim W((\mathbf{S}\nu)^{-1}, \nu)$, then this represents conjugacy, since the following result can be derived:

$$\Sigma_x^{-1}|\mathbf{x}_1, \dots, \mathbf{x}_T \sim W\left(\left(\sum_{i=1}^T (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})' + \nu\mathbf{S}\right)^{-1}, \nu + T\right), \quad (8.46)$$

where recall that $W(\cdot, \cdot)$ denotes a Wishart distribution.

Using (8.45), we can obtain the full conditional distributions for the state process as follows:

$$\mathbf{Y}_0|\cdot \sim \text{Gau}(\mathbf{V}_0\mathbf{a}_0, \mathbf{V}_0), \quad (8.47)$$

where

$$\begin{aligned} \mathbf{V}_0 &\equiv (\mathbf{M}'\mathbf{Q}^{-1}\mathbf{M} + \Sigma_0^{-1})^{-1}, \\ \mathbf{a}_0 &\equiv \mathbf{M}'\mathbf{Q}^{-1}\mathbf{Y}_1 + \Sigma_0^{-1}\boldsymbol{\mu}_0; \end{aligned}$$

similarly, for $t = 1, \dots, T-1$, we have

$$\mathbf{Y}_t|\cdot \sim \text{Gau}(\mathbf{V}_t\mathbf{a}_t, \mathbf{V}_t), \quad (8.48)$$

where

$$\begin{aligned} \mathbf{V}_t &\equiv (\mathbf{H}_t'\mathbf{R}^{-1}\mathbf{H}_t + \mathbf{Q}^{-1} + \mathbf{M}'\mathbf{Q}^{-1}\mathbf{M})^{-1}, \\ \mathbf{a}_t &\equiv \mathbf{H}_t'\mathbf{R}^{-1}\mathbf{Z}_t + \mathbf{Q}^{-1}\mathbf{M}\mathbf{Y}_{t-1} + \mathbf{M}'\mathbf{Q}^{-1}\mathbf{Y}_{t+1}; \end{aligned}$$

and for $t = T$,

$$\mathbf{Y}_T|\cdot \sim \text{Gau}(\mathbf{V}_T\mathbf{a}_T, \mathbf{V}_T), \quad (8.49)$$

where

$$\begin{aligned} \mathbf{V}_T &\equiv (\mathbf{H}_T'\mathbf{R}^{-1}\mathbf{H}_T + \mathbf{Q}^{-1})^{-1}, \\ \mathbf{a}_T &\equiv \mathbf{H}_T'\mathbf{R}^{-1}\mathbf{Z}_T + \mathbf{Q}^{-1}\mathbf{M}\mathbf{Y}_{T-1}. \end{aligned}$$

The full conditional distributions of the covariance matrices can be obtained by using (8.46). Thus,

$$\mathbf{R}^{-1} | \cdot \sim W \left(\left(\sum_{t=1}^T (\mathbf{Z}_t - \mathbf{H}_t \mathbf{Y}_t)(\mathbf{Z}_t - \mathbf{H}_t \mathbf{Y}_t)' + \nu_R \mathbf{C}_R \right)^{-1}, \nu_R + T \right), \quad (8.50)$$

$$\mathbf{Q}^{-1} | \cdot \sim W \left(\left(\sum_{t=1}^T (\mathbf{Y}_t - \mathbf{M} \mathbf{Y}_{t-1})(\mathbf{Y}_t - \mathbf{M} \mathbf{Y}_{t-1})' + \nu_Q \mathbf{C}_Q \right)^{-1}, \nu_Q + T \right). \quad (8.51)$$

The full conditional distribution for $\mathbf{M}$ also makes use of the Gaussian/Gaussian result (8.45), but it is facilitated by rewriting the evolution model (8.39) in a matrix format, as follows. Define $\mathcal{Y}_{1:T} \equiv [\mathbf{Y}_1, \dots, \mathbf{Y}_T]$, an $n \times T$ matrix. Similarly, define the $n \times T$ matrices, $\mathcal{Y}_{0:T-1} \equiv [\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_{T-1}]$ and $\mathcal{U}_{1:T} \equiv [\boldsymbol{\eta}_1, \dots, \boldsymbol{\eta}_T]$. Thus, we can rewrite the evolution model (8.39) as $\mathcal{Y}_{1:T} = \mathbf{M} \mathcal{Y}_{0:T-1} + \mathcal{U}_{1:T}$. This, in turn, can be written equivalently as

$$\text{vec}(\mathcal{Y}_{1:T}) = (\mathcal{Y}'_{0:T-1} \otimes \mathbf{I}_n) \mathbf{m} + \text{vec}(\mathcal{U}_{1:T}), \quad (8.52)$$

where $\text{var}(\text{vec}(\mathcal{U}_{1:T})) = \mathbf{I}_T \otimes \mathbf{Q} \equiv \tilde{\mathbf{Q}}$, and recall that $\mathbf{m} \equiv \text{vec}(\mathbf{M})$. Now, using (8.43) and (8.52) as well as the Gaussian/Gaussian result (8.45), we obtain the full conditional distribution,

$$\mathbf{m} | \cdot \sim \text{Gau}(\mathbf{V}_m \mathbf{a}_m, \mathbf{V}_m), \quad (8.53)$$

where

$$\begin{aligned} \mathbf{V}_m &\equiv ((\mathcal{Y}'_{0:T-1} \otimes \mathbf{I}_n)' \tilde{\mathbf{Q}}^{-1} (\mathcal{Y}'_{0:T-1} \otimes \mathbf{I}_n) + \boldsymbol{\Sigma}_m^{-1})^{-1}, \\ \mathbf{a}_m &\equiv (\mathcal{Y}'_{0:T-1} \otimes \mathbf{I}_n)' \tilde{\mathbf{Q}}^{-1} \text{vec}(\mathcal{Y}_{1:T}) + \boldsymbol{\Sigma}_m^{-1} \boldsymbol{\mu}_m. \end{aligned} \quad (8.54)$$

The *Gibbs sampler algorithm* can be written in pseudo code as follows:

Select hyperparameters: $\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0, \nu_R, \mathbf{C}_R, \nu_Q, \mathbf{C}_Q, \boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m$

Select initial values: $\{\mathbf{Y}_t^{(0)} : t = 0, \dots, T\}, \mathbf{M}^{(0)}, \mathbf{R}^{(0)}, \mathbf{Q}^{(0)}$

For $i = 1$ to N_{gibbs}

- Sample from $\mathbf{Y}_0^{(i)} | \mathbf{Y}_1^{(i-1)}, \mathbf{M}^{(i-1)}, \mathbf{Q}^{(i-1)}$, using (8.47).
- For $t = 1, \dots, T - 1$, sample from $\mathbf{Y}_t^{(i)} | \mathbf{Y}_{t-1}^{(i)}, \mathbf{Y}_{t+1}^{(i-1)}, \mathbf{M}^{(i-1)}, \mathbf{Q}^{(i-1)}$, using (8.48).
- Sample from $\mathbf{Y}_T^{(i)} | \mathbf{Y}_{T-1}^{(i)}, \mathbf{M}^{(i-1)}, \mathbf{Q}^{(i-1)}$, using (8.49).
- Sample from $\mathbf{R}^{(i)} | \{\mathbf{Y}_t^{(i)} : t = 0, \dots, T\}$, using (8.50).
- Sample from $\mathbf{Q}^{(i)} | \{\mathbf{Y}_t^{(i)} : t = 0, \dots, T\}, \mathbf{M}^{(i-1)}$, using (8.51).
- Sample from $\mathbf{m}^{(i)} | \{\mathbf{Y}_t^{(i)} : t = 0, \dots, T\}, \mathbf{Q}^{(i)}$, using (8.53)

end

The Gibbs sampler algorithm for the BHM presented in this section can be made more efficient, in the sense of faster convergence, if we sample the state process jointly, rather than individually. We saw in Section 8.2.1 that the basis of the Kalman filter/smoothing algorithm gives us the ability to obtain sequentially the joint distribution of the state process at all times, given all observations. Effectively, we can use the Kalman filter algorithms within the Gibbs iterations to obtain the joint distribution of the states as follows (e.g., Shumway and Stoffer, 2006, Section 6.9). First, note that from the Markov assumption, we have

$$\begin{aligned} [\mathbf{Y}_0, \dots, \mathbf{Y}_T | \mathbf{Z}_{1:T}, \Theta] &= [\mathbf{Y}_0 | \mathbf{Y}_1, \Theta][\mathbf{Y}_1 | \mathbf{Y}_2, \mathbf{Z}_{1:1}, \Theta] \\ &\dots [\mathbf{Y}_{T-1} | \mathbf{Y}_T, \mathbf{Z}_{1:T}, \Theta][\mathbf{Y}_T | \mathbf{Z}_{1:T}, \Theta], \end{aligned} \quad (8.55)$$

where here Θ denotes the parameter matrices $\mathbf{M}$, $\mathbf{R}$, and $\mathbf{Q}$. Thus, starting from the last distribution on the right-hand side of (8.55), we could sample $\mathbf{Y}_T$, and then we could use that to sample $\mathbf{Y}_{T-1}$ from the second-to-last distribution on the right-hand side of (8.55), and so on, until we obtain samples for all $\{\mathbf{Y}_t : t = T, \dots, 0\}$. To do this, we must be able to sample easily from $[\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}, \Theta]$, for $t = T-1, \dots, 0$. Fortunately, this is feasible by making use of results from the Kalman filter/smoothing. First, recall from (8.10) that

$$[\mathbf{Y}_t | \mathbf{Y}_{t+1}, \mathbf{Z}_{1:t}, \Theta] \propto [\mathbf{Y}_{t+1} | \mathbf{Y}_t, \mathbf{Z}_{1:t}, \Theta][\mathbf{Y}_t | \mathbf{Z}_{1:T}], \quad (8.56)$$

and if the two components are Gaussian distributions, this is precisely (8.25). By completing the square, the distribution is Gaussian with mean and variance given by (8.26) and (8.27), respectively. This can be used in the Gibbs sampler in what is known as the *forward filtering, backward sampling algorithm* (Carter and Kohn, 1994; Frühwirth-Schnatter, 1994). Specifically, rather than sample $\{\mathbf{Y}_t : t = 0, \dots, T\}$ individually in the Gibbs sampler, we would like to sample $[\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_T | \cdot]$, which is accomplished by carrying out the following steps:

1. Run the Kalman filter with the current Gibbs samples for Θ substituted into $\mathbf{Y}_{t|t}$, $\mathbf{Y}_{t|t-1}$, $\mathbf{P}_{t|t}$, and $\mathbf{P}_{t|t-1}$.
2. Sample $\mathbf{Y}_T^{(i)}$ from $\text{Gau}(\mathbf{Y}_{T|T}, \mathbf{P}_{T|T})$.
3. For $t = T-1, \dots, 0$, sample $\mathbf{Y}_t^{(i)}$ from $\text{Gau}(\mathbf{Y}_{t|t} + \mathbf{J}_t(\mathbf{Y}_{t+1}^{(i)} - \mathbf{Y}_{t+1|t}), \mathbf{P}_{t|t} - \mathbf{J}_t\mathbf{P}_{t+1|t}\mathbf{J}_t')$, using the most recently sampled $\mathbf{Y}_{t+1}^{(i)}$.

Although using the forward filtering, backward sampling algorithm may seem excessively complicated compared to the basic Gibbs sampler presented previously, typically it produces faster convergence in the sampler if the dependence between $\mathbf{Y}_t$ and $\mathbf{Y}_{t-1}$ is moderate to strong.

Example: As a simple illustration of the Gibbs-sampler algorithm described above, consider the case of simulating noisy data from the AR(1) process given in Section 8.2.1; the data model is given by (8.23) and the process model is given by (8.24). We also specify a prior for the initial condition, namely $Y_0 \sim \text{Gau}(0, 1)$. For this illustration, we assume that we “know” the measurement-error variance (i.e., $R = 0.1$), the evolution operator M has prior distribution, $M \sim \text{Unif}(-1, 1)$, and the process-model variance has prior distribution, $\sigma_\eta^2 \sim \text{IG}(2, 1)$. [The notation, $\text{Unif}(u_1, u_2)$, corresponds to a continuous uniform distribution between u_1 and u_2 , and $\text{IG}(a, b)$ corresponds to an inverse gamma distribution with shape parameter a and scale parameter b .] The MCMC in this case seeks the posterior distributions of M , σ_η^2 , and $Y_0, Y_1, \dots, Y_T$. Then the Gibbs-sampler algorithm is a special case of the more general (matrix/vector) algorithm presented above. Specific formulas for the full-conditional distributions for this example are given in Wikle and Berliner (2007, Section 5.3.1). The algorithm was implemented on the simulated data shown in Figure 8.1, with an MCMC sample of 45,000 after a burn-in period of 5000 iterations. Figure 8.4 shows a histogram of samples from the posterior distributions of M and σ_η^2 . Recall that the true values for these parameters used in the simulation were $M = 0.7$ and $\sigma_\eta^2 = 0.5$; clearly, the posterior distributions cover these true values. Figure 8.5 shows the posterior mean of the state process $\{Y_t\}$ as well as the 2.5 percentiles and 97.5 percentiles from the posterior distribution of the state process for each time, which is a way to

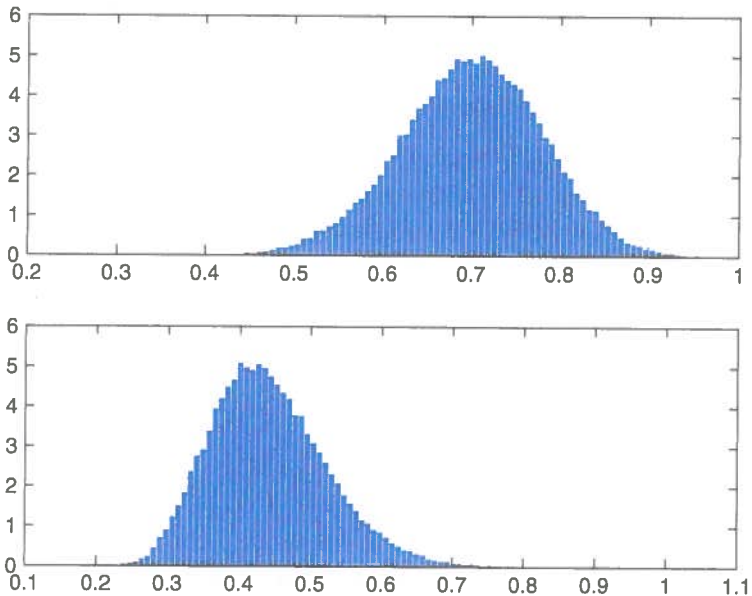


Figure 8.4 Histogram of MCMC samples from the posterior distribution of the parameters. **Top panel:** Evolution operator, M . **Bottom panel:** Process-model variance, σ_η^2 .

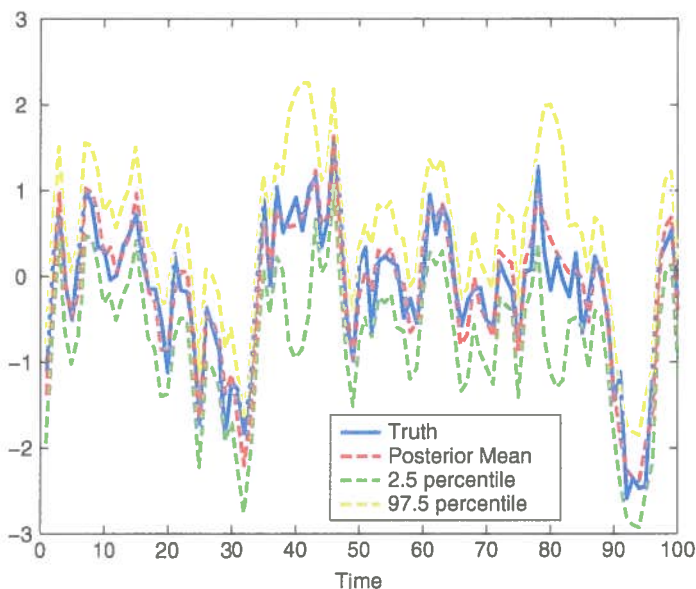


Figure 8.5 Summary of MCMC output for the posterior distribution of the state process, $\{Y_t\}$. Solid blue line: “truth”; dashed red line: posterior mean; dashed green line: lower 2.5 percentiles; dash-dot green line: upper 97.5 percentiles.

measure uncertainty in the posterior distribution. Observe that the posterior distribution generally covers the “truth,” and the uncertainty in the time periods with no data (at times 40–43 and 80–83) is much larger than in periods where there are data, as expected.

Although Gaussian linear DSTMs are fairly simple to implement in principle, there can be practical computational issues when the dimension of the state process and/or parameter vectors is high. Projecting these vectors onto lower dimensional manifolds or basis function expansions as described in Chapter 7 can help ease the computational burden. In addition, one can exploit simplifying reparameterizations that arise from matrix decompositions of the parameter matrices, such as the Cholesky decomposition (e.g., Eubank 2006; Lee and Ghosh 2008).

8.4 INFERENCE FOR THE HIERARCHICAL DSTM: NONLINEAR/NON-GAUSSIAN MODELS

The Kalman filter algorithm and the Kalman smoother algorithm are most appropriate for state-space models with linear operators and Gaussian error. Unfortunately, many spatio-temporal statistical models violate these assumptions. In this section, we discuss several options for accommodating

more complicated spatio-temporal structures. We refer the reader to the Bibliographic Notes in Section 8.5 for references that give more complete coverage.

8.4.1 Extended Kalman Filter

Consider the general DSTM formulation within an HM: The data model is

$$\mathbf{Z}_t = \mathcal{H}_t(\mathbf{Y}_t; \boldsymbol{\theta}_D) + \boldsymbol{\varepsilon}_t, \quad (8.57)$$

and the process model is

$$\mathbf{Y}_t = \mathcal{M}_t(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_P) + \boldsymbol{\eta}_t, \quad (8.58)$$

where $\mathcal{H}_t(\cdot)$ and $\mathcal{M}_t(\cdot)$ are known, nonlinear functions. While linear and additive combinations of Gaussian random variables have Gaussian distributions, this is not the case for nonlinear transformations. Thus, it is not necessary that the error terms in the data model and the process model be Gaussian—or even additive—although they are additive in (8.57) and (8.58). In general, the errors should be included in the functions $\mathcal{H}_t$ and $\mathcal{M}_t$. However, in the context described here, one assumes that the first two moments are sufficient to describe the state and measurement distributions, and thus additive error terms are often assumed. In this section, we assume that the errors $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\eta}_t$ in (8.57) and (8.58), respectively, have mean zero and are temporally independent within and between series.

The traditional approach to this problem when equations (8.57) and (8.58) are not far from linear is to employ some sort of linearization (e.g., West and Harrison, 1997, Section 13.2; Grewal and Andrews, 1993, Chapter 5). Primarily, this involves *tangent-linear expansions* of $\mathcal{H}_t$ and $\mathcal{M}_t$. That is, one expands these functions using Taylor series and then truncates, typically considering terms only up to first order (as discussed in Sections 7.1.4 and 7.3.1). Thus, the nonlinear functions are approximated by

$$\mathcal{H}_t \simeq \left[\frac{\partial \mathcal{H}_t(\mathbf{Y}_t)}{\partial \mathbf{Y}_t} \right]_{\mathbf{Y}_t = \hat{\mathbf{Y}}_t} \equiv \tilde{\mathbf{H}}_t, \quad (8.59)$$

$$\mathcal{M}_{t-1} \simeq \left[\frac{\partial \mathcal{M}_t(\mathbf{Y}_{t-1})}{\partial \mathbf{Y}_{t-1}} \right]_{\mathbf{Y}_{t-1} = \hat{\mathbf{Y}}_{t-1}} \equiv \tilde{\mathbf{M}}_{t-1}, \quad (8.60)$$

where $\{\hat{\mathbf{Y}}_t\}$ are suitable state estimates about which to linearize. A common choice is the forecasted trajectory of the state process. These linear approximations are then used in the “update step” to update the filter’s state vector and covariance matrices; however, the full nonlinear equations are used in the forecast step. Given the measurement and evolution error covariance matrices, $\mathbf{R}_t$ and $\mathbf{Q}_t$, respectively, and initial values $\mathbf{Y}_{0|0}$ and $\mathbf{P}_{0|0}$, the algorithm, often referred to as the *extended Kalman filter* (EKF), can be written in pseudo code as follows:

For $t = 1$ to T

1. Obtain the forecast distribution's mean and covariances:
 - (a) $\mathbf{Y}_{t|t-1} = \mathcal{M}_t(\mathbf{Y}_{t-1|t-1})$.
 - (b) $\mathbf{P}_{t|t-1} = \tilde{\mathbf{M}}_{t-1}\mathbf{P}_{t-1|t-1}\tilde{\mathbf{M}}'_{t-1} + \mathbf{Q}_{t-1}$.
2. Calculate the gain matrix, $\mathbf{K}_t = \mathbf{P}_{t|t-1}\tilde{\mathbf{H}}'_t(\tilde{\mathbf{H}}'_t\mathbf{P}_{t|t-1}\tilde{\mathbf{H}}_t + \mathbf{R}_t)^{-1}$.
3. Obtain the filter distribution's mean and covariances:
 - (a) $\mathbf{Y}_{t|t} = \mathbf{Y}_{t|t-1} + \mathbf{K}_t(\mathbf{Z}_t - \mathcal{H}_t(\mathbf{Y}_{t|t-1}))$.
 - (b) $\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\tilde{\mathbf{H}}_t)\mathbf{P}_{t|t-1}$.

end

Note that one uses the current EKF forecast value for $\hat{\mathbf{Y}}_t$ in this algorithm.

8.4.2 BHM Implementation via MCMC

In principle, the BHM formulation can accommodate nonlinear and non-Gaussian structures. This can be implemented through MCMC or through *importance sampling*. Importance sampling will be discussed in the next subsection. Here, we briefly discuss MCMC-related implementation issues, following Carlin, Polson, and Stoffer (1992) and Shumway and Stoffer (2006, Chapter 6). Note that implementation issues for non-Gaussian data models are typically similar to those found in spatial statistics (e.g., Banerjee, Carlin, and Gelfand, 2004; Diggle and Ribeiro, 2007), since the observations are usually assumed conditionally independent, given the latent spatio-temporal dynamical process.

To illustrate the basic ideas of MCMC for a BHM, we consider several cases. Consider the case where the data model is linear and the evolution model is nonlinear:

$$\mathbf{Z}_t = \mathbf{H}_t\mathbf{Y}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{R}), \quad (8.61)$$

$$\mathbf{Y}_t = \mathcal{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_P) + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim iid \text{Gau}(\mathbf{0}, \mathbf{Q}). \quad (8.62)$$

Now, assume that the Gibbs sampler is such that we need to sample from the full conditional distribution, $[\mathbf{Y}_t | \cdot]$. Thus,

$$\begin{aligned} [\mathbf{Y}_t | \cdot] &\propto [\mathbf{Z}_t | \mathbf{Y}_t][\mathbf{Y}_t | \mathbf{Y}_{t-1}][\mathbf{Y}_{t+1} | \mathbf{Y}_t] \\ &\propto \text{Gau}(\mathbf{A}_{1t}\mathbf{a}_{1t}, \mathbf{A}_{1t})[\mathbf{Y}_{t+1} | \mathbf{Y}_t], \end{aligned}$$

where once again there is a slight abuse of notation in writing $\text{Gau}(\cdot, \cdot)$ instead of its density, and

$$\begin{aligned} \mathbf{A}_{1t} &\equiv (\mathbf{Q}^{-1} + \mathbf{H}'_t\mathbf{R}^{-1}\mathbf{H}_t)^{-1}, \\ \mathbf{a}_{1t} &\equiv \mathbf{Q}^{-1}\mathcal{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_P) + \mathbf{H}'_t\mathbf{R}^{-1}\mathbf{Z}_t. \end{aligned}$$

However, note that

$$\begin{aligned} [\mathbf{Y}_{t+1} | \mathbf{Y}_t] &\propto \exp\{-(1/2)[(\mathbf{Y}_{t+1} - \mathcal{M}(\mathbf{Y}_t; \boldsymbol{\theta}_P))' \mathbf{Q}^{-1} (\mathbf{Y}_{t+1} - \mathcal{M}(\mathbf{Y}_t; \boldsymbol{\theta}_P))]\} \\ &\equiv \gamma_1(\mathbf{Y}_t), \end{aligned} \quad (8.63)$$

and since $\mathcal{M}(\mathbf{Y}_t; \boldsymbol{\theta}_P)$ is nonlinear in $\mathbf{Y}_t$ (and $\boldsymbol{\theta}_P$), the full conditional distribution cannot be obtained analytically. One could use a Metropolis–Hastings step to sample this full conditional distribution. The key to using a Metropolis–Hastings step within a Gibbs sampler is to find a good proposal distribution. One such proposal distribution, when the evolution equation is not far from linear, involves using a linearized model distribution in place of $\gamma_1(\mathbf{Y}_t)$, such as suggested by the EKF. By making this Gaussian, the Gaussian/Gaussian result (8.45) can be used to derive an approximate full conditional distribution that is Gaussian. The proposal in the Metropolis–Hastings step is then obtained by sampling from this approximate full conditional distribution. Of course, in high dimensions, the efficiency of this algorithm can be quite low.

Alternatively, as discussed in Shumway and Stoffer (2006, Section 6.9), because $0 \leq \gamma_1(\mathbf{Y}_t) \leq 1$, for all $\mathbf{Y}_t$, the distribution from which we wish to sample is controlled by $\text{Gau}(\mathbf{A}_{1t} \mathbf{a}_{1t}, \mathbf{A}_{1t})$, and it is reasonable to use *rejection sampling* (e.g., see Robert and Casella, 2004, Chapter 2) to sample from the desired full conditional distribution. Specifically, we generate $\mathbf{Y}_t^{(i)}$ from $\text{Gau}(\mathbf{A}_{1t} \mathbf{a}_{1t}, \mathbf{A}_{1t})$, and we accept it with probability $\gamma_1(\mathbf{Y}_t)$. For $\gamma_1(\mathbf{Y}_t)$ close to zero, rejection sampling will be inefficient, and alternatives, such as adaptive-rejection sampling (e.g., see Robert and Casella, 2004, Chapter 2), should be considered. Often, in the DSTM setting, n is large, and it is thus the case that $\gamma_1(\mathbf{Y}_t)$ is quite small, in which case even adaptive rejection sampling can be very inefficient. Furthermore, in the typical case where the parameters $\boldsymbol{\theta}_P$ are not known *a priori*, they must be sampled as well in the Gibbs sampler. In that case, a Metropolis–Hastings step is typically used.

Now, consider the case where the data model is nonlinear and the evolution model is linear:

$$\mathbf{Z}_t = \mathcal{H}(\mathbf{Y}_t; \boldsymbol{\theta}_D) + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \text{iid } \text{Gau}(\mathbf{0}, \mathbf{R}), \quad (8.64)$$

$$\mathbf{Y}_t = \mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \text{iid } \text{Gau}(\mathbf{0}, \mathbf{Q}). \quad (8.65)$$

Analogous to the previous case, the full conditional distribution for $\mathbf{Y}_t$ can be written as

$$[\mathbf{Y}_t | \cdot] \propto \gamma_2(\mathbf{Y}_t) \text{Gau}(\mathbf{A}_{2t} \mathbf{a}_{2t}, \mathbf{A}_{2t}),$$

where again there is a slight abuse of notation,

$$\begin{aligned} \mathbf{A}_{2t} &\equiv (\mathbf{Q}^{-1} + \mathbf{M}'\mathbf{Q}^{-1}\mathbf{M})^{-1}, \\ \mathbf{a}_{2t} &\equiv \mathbf{Q}^{-1}\mathbf{M}\mathbf{Y}_{t-1} + \mathbf{M}'\mathbf{Q}^{-1}\mathbf{Y}_{t+1}, \end{aligned}$$

and

$$\gamma_2(\mathbf{Y}_t) \equiv \exp\{-(1/2)(\mathbf{Z}_t - \mathcal{H}(\mathbf{Y}_t; \boldsymbol{\theta}_D))' \mathbf{R}^{-1} (\mathbf{Z}_t - \mathcal{H}(\mathbf{Y}_t; \boldsymbol{\theta}_D))\}.$$

Again, the full conditional distribution cannot be obtained analytically. However, we can use a linearized distribution in place of $\gamma_2(\mathbf{Y}_t)$ in a Metropolis–Hastings step; or we could use a rejection-sampling approach, by sampling from $Gau(\mathbf{A}_{2t}, \mathbf{a}_{2t}, \mathbf{A}_{2t})$ and accepting the sample with probability $\gamma_2(\mathbf{Y}_t)$. Again, if $\gamma_2(\mathbf{Y}_t)$ is close to zero (as often happens), this will be inefficient and one must use alternative sampling algorithms (e.g., adaptive-rejection, Metropolis–Hastings, or slice sampling).

Finally, consider the case where both the data model and the evolution model are nonlinear, as given by (8.57) and (8.58), where $\{\boldsymbol{\varepsilon}_t\}$ and $\{\boldsymbol{\eta}_t\}$ are assumed Gaussian. Then the full conditional distribution is

$$[\mathbf{Y}_t | \cdot] \propto \gamma_1(\mathbf{Y}_t) \gamma_2(\mathbf{Y}_t) Gau(\mathcal{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_P), \mathbf{Q}).$$

As before, this suggests that one can use rejection sampling to sample from this distribution; that is, first sample from $Gau(\mathcal{M}(\mathbf{Y}_{t-1}; \boldsymbol{\theta}_P), \mathbf{Q})$, and then accept it with probability $\gamma_1(\mathbf{Y}_t) \gamma_2(\mathbf{Y}_t)$. Such an approach is often inefficient, since $\gamma_1(\mathbf{Y}_t) \gamma_2(\mathbf{Y}_t)$ is typically very small. Then, adaptive-rejection sampling or a Metropolis–Hastings step is often used instead.

8.4.3 Alternative Metropolis–Hastings Algorithms and Approximations

Unfortunately, Metropolis–Hastings updates for high-dimensional nonlinear state-space models with unknown parameters can be difficult to implement and typically require extensive user-informed tuning. Thus, in the spirit of effective model/computation trade-offs (Section 2.5), one might choose to approximate the first-order Markovian dependence using a second-order temporal-dependence model that avoids computational difficulties. For example, the MCMC algorithm employed to fit the GQN ecological reaction-diffusion model described by Hooten and Wikle (2008) contains an approximation to a quadratic evolution equation, where the second-order term, $\boldsymbol{\theta} \cdot \text{diag}(\mathbf{Y}_{t-1}) \mathbf{Y}_{t-2}$, is used rather than $\boldsymbol{\theta} \cdot \text{diag}(\mathbf{Y}_{t-1}) \mathbf{Y}_{t-1}$ (and $\boldsymbol{\theta}$ is a process-model parameter). With Gaussian error assumptions, this slight change in the evolution equation to a second-order equation allows analytical full conditional distributions to be obtained for the state variable $\mathbf{Y}_t$, resulting in more efficient computation via Gibbs sampling. Of course, in many cases, one would not want to change the dependence structure in the process model, but this approach could still be used to generate useful proposals in a block Metropolis-update algorithm. Under the GQN framework discussed in Section 7.3.3, such modifications are only possible when the function $g(\cdot)$ is linear, and thus they are only applicable in specific situations.

Note that many other approximations have been proposed so that the efficiency and robustness of the Gibbs sampler can be assured in the implementation of nonlinear models (e.g., Wakefield et al., 1994), sometimes through additional rejection algorithms (e.g., Carlin, Polson, and Stoffer, 1992; Jungbacker and Koopman, 2007). There has been considerable recent work to develop efficient MCMC algorithms for general, high-dimensional, highly nonlinear models. These include the “differential evolution MCMC” approach of ter Braak (2006), the “delayed rejection adaptive Metropolis” (DRAM) sampling algorithm of Haario et al. (2006), the “multiple very fast simulated annealing” (MVFSa) algorithm of Villagran et al. (2008), the “differential evolution adaptive Metropolis” (DREAM) algorithm of Vrugt et al. (2009), and the “t-walk” general-purpose MCMC sampler of Christen and Fox (2010). All of these methods show promise for increasing efficiency over standard MCMC algorithms for complex model implementations. However, as discussed in the overview of Higdon et al. (2008), although these algorithms (and other similar ones) can be quite effective in improving convergence for nonlinear spatial and spatio-temporal processes, in some cases, relatively simple single-component Metropolis updates can be as efficient as these more complicated sampling algorithms. At this point, our best advice is to evaluate a number of algorithms to see which is best for a specific application.

8.4.4 Importance Sampling Monte Carlo

Importance sampling Monte Carlo is another approach that can be used with a BHM (or an EHM) in the case where models are nonlinear or non-Gaussian (e.g., Robert and Casella, 2004, Section 3.3). We present here a general overview of importance sampling, followed by a special case that is useful for DSTMs, known as sequential importance sampling.

Historically, the primary use of Monte Carlo (MC) has been for the estimation of integrals (i.e., expectations for given probability models). Let g be a possibly multivariate function of the state process, $\mathbf{Y}_0, \dots, \mathbf{Y}_t$; recall that $\mathbf{Z}_{1:t}$ denotes all available data up to and including time t ; and suppose we are interested in the conditional expectation,

$$\begin{aligned} E(g(\mathbf{Y}_{0:t})|\mathbf{Z}_{1:t}) &= \int g(\mathbf{Y}_{0:t})[\mathbf{Y}_{0:t}|\mathbf{Z}_{1:t}]d\mathbf{Y}_{0:t} \\ &= \frac{\int g(\mathbf{Y}_{0:t})[\mathbf{Z}_{1:t}|\mathbf{Y}_{0:t}][\mathbf{Y}_{0:t}]d\mathbf{Y}_{0:t}}{\int [\mathbf{Z}_{1:t}|\mathbf{Y}_{0:t}][\mathbf{Y}_{0:t}]d\mathbf{Y}_{0:t}}, \end{aligned}$$

assuming the integrals exist. An MC estimate can be obtained through the following *MC integration algorithm* (written in pseudo code):

1. Generate L (pseudo-random) realizations, $\mathbf{Y}_{0:t}^{(l)}$, from $[\mathbf{Y}_{0:t}|\mathbf{Z}_{1:t}]$; $l = 1, \dots, L$.

2. Evaluate g for each realization and compute the arithmetic average of the results, $\hat{E}(g(\mathbf{Y}_{0:t})|\mathbf{Z}_{1:t}) \equiv (1/L) \sum_{l=1}^L g(\mathbf{Y}_{0:t}^{(l)})$.

Under independent sampling in 1., this average converges (almost surely) to $E(g(\mathbf{Y}_{0:t})|\mathbf{Z}_{1:t})$ as L tends to infinity, which is a result of the strong law of large numbers. This also holds if realizations are stationary and ergodic, though not necessarily independent (Section 2.3).

We can also approximate the posterior probability density, $p(\cdot|\mathbf{Z}_{1:t}) \equiv [Y_{0:t}|\mathbf{Z}_{1:t}]$ as follows. Let $\mathbf{y}_{0:t}$ denote a generic argument of $p(\cdot|\mathbf{Z}_{1:t})$. Then define $g(\mathbf{y})$ according to the scaled indicator function,

$$g(\mathbf{y}) \equiv \frac{I(\mathbf{y} \in (\mathbf{y}_{0:t}, \mathbf{y}_{0:t} + \Delta))}{|\Delta_0||\Delta_1| \dots |\Delta_t|},$$

where $\Delta \equiv (\Delta_0, \Delta_1, \dots, \Delta_t)$ is a $(t+1)$ -dimensional vector whose components are all nonzero. Define

$$\hat{p}(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t}) \equiv \hat{E}(g(\mathbf{Y}_{0:t})|\mathbf{Z}_{1:t}) = \frac{1}{L} \sum_{l=1}^L g(\mathbf{Y}_{0:t}^{(l)}),$$

which is a good approximation to $p(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t})$ as both $\max\{|\Delta_i|\} \rightarrow 0$ and $L \rightarrow \infty$. In practice, one typically considers kernel-density estimates based on the realizations $\{\mathbf{Y}_{0:t}^{(l)} : l = 1, \dots, L\}$.

Notice that we have introduced the notation, $p(\cdot|\mathbf{Z}_{1:t})$, to emphasize the functional form of the posterior probability density. This notation is also useful when we sample from another density that is not the posterior density. When direct simulation from $p(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t})$ is difficult or impossible, we can use *importance sampling Monte Carlo (ISMC)*. The idea is to consider another density with the same support as $p(\cdot|\mathbf{Z}_{1:t})$, say $q(\cdot|\mathbf{Z}_{1:t})$, that is comparatively easy to sample from. Then, we generate L samples $\{\mathbf{Y}_{0:t}^{(l)} : l = 1, \dots, L\}$ from the *importance density*, $q(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t})$, and evaluate $g(\mathbf{Y}_{0:t}^{(l)})$ for each $l = 1, \dots, L$. However, these samples are not from $p(\cdot|\mathbf{Z}_{1:t})$, so to use these to estimate $E(g(\mathbf{Y}_{0:t})|\mathbf{Z}_{1:t})$, each sample (ensemble) member must be weighted. That is, the estimate is

$$\hat{E}(g(\mathbf{Y}_{0:t})|\mathbf{Z}_{1:t}) \equiv \sum_{l=1}^L w_t^{(l)} g(\mathbf{Y}_{0:t}^{(l)}),$$

where the ISMC weights, $\{w_t^{(l)} : l = 1, \dots, L\}$, are given below.

From the ISMC perspective, we have

$$E(g(\mathbf{Y}_{0:t})|\mathbf{Z}_{1:t}) = \int g(\mathbf{y}_{0:t}) \frac{p(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t})}{q(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t})} q(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t}) d\mathbf{y}_{0:t}. \quad (8.66)$$

Then $v_t^{(l)}$, the unnormalized weight associated with $\mathbf{Y}_{0:t}^{(l)}$, is given by $p(\mathbf{Y}_{0:t}^{(l)}|\mathbf{Z}_{1:t})/q(\mathbf{Y}_{0:t}^{(l)}|\mathbf{Z}_{1:t})$, $l = 1, \dots, L$. An intuitive choice for the importance density q in this case is the “prior,” or evolution distribution,

$$q(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t}) \equiv p(\mathbf{y}_{0:t}) = p(\mathbf{y}_0) \prod_{k=1}^t p(\mathbf{y}_k|\mathbf{y}_{k-1}),$$

which, in fact, does not depend on the data. Thus, given samples from $p(\mathbf{y}_0)$, we could obtain MC samples for all times $k = 1, \dots, t$, using the evolution model. In this case, the unnormalized weights are just given by the likelihood, since

$$\frac{p(\mathbf{Y}_{0:t}|\mathbf{Z}_{1:t})}{p(\mathbf{Y}_{0:t})} \propto \frac{p(\mathbf{Z}_{1:t}|\mathbf{Y}_{0:t})p(\mathbf{Y}_{0:t})}{p(\mathbf{Y}_{0:t})} = p(\mathbf{Z}_{1:t}|\mathbf{Y}_{0:t}).$$

That is, for a sample, $\mathbf{Y}_{0:t}^{(l)} \equiv \{\mathbf{Y}_0^{(l)}, \mathbf{Y}_1^{(l)}, \dots, \mathbf{Y}_t^{(l)}\}$, $l = 1, \dots, L$, generated from the prior, an estimate for $p(\cdot|\mathbf{Z}_{1:T})$ can be obtained by using the *normalized* ISMC weights,

$$w_t^{(l)} \equiv \frac{\prod_{k=1}^t p(\mathbf{Z}_k|\mathbf{Y}_k^{(l)})}{\sum_{j=1}^L \prod_{k=1}^t p(\mathbf{Z}_k|\mathbf{Y}_k^{(j)})}, \quad l = 1, \dots, L, \quad (8.67)$$

where conditional independence, $p(\mathbf{Z}_{1:t}|\mathbf{Y}_{0:t}) = \prod_{k=1}^t p(\mathbf{Z}_k|\mathbf{Y}_k)$, is a consequence of (8.2). Specifically, $E(g(\mathbf{Y}_{0:t}|\mathbf{Z}_{1:t}))$ can be approximated by $\sum_{l=1}^L w_t^{(l)} g(\mathbf{Y}_{0:t}^{(l)})$, and the posterior density $p(\cdot|\mathbf{Z}_{1:t})$ can be approximated by applying a *weighted* kernel density estimate (with weights $\{w_t^{(l)}: l = 1, \dots, L\}$) to the sample $\{\mathbf{Y}_{0:t}^{(1)}, \dots, \mathbf{Y}_{0:t}^{(L)}\}$ from the prior.

Now put $t = T$. Then this ensemble-based approach for obtaining samples from the posterior distribution of $\mathbf{Y}_{0:T}$ given $\mathbf{Z}_{1:T}$, is sometimes referred to as an *ensemble smoother*.

Although, in principle, this approach is sound, there are practical issues that limit its use in high dimensions. In particular, as discussed in Berliner and Wikle (2007), consider the case presented here where the importance distribution is the prior and the ISMC weights are proportional to the likelihoods [i.e., equation (8.67)]. Consider a K -variate random vector $\mathbf{d}$ whose distribution is $\text{Gau}(\mathbf{a}, \Sigma)$, with density written as $p(\mathbf{d}|\mathbf{a}) = c \cdot \exp\{-(1/2)(\mathbf{d} - \mathbf{a})'\Sigma^{-1}(\mathbf{d} - \mathbf{a})\}$, where c is the normalizing constant. From probability theory, we know that the quadratic form, $(\mathbf{d} - \mathbf{a})'\Sigma^{-1}(\mathbf{d} - \mathbf{a})$, has a chi-squared distribution on K degrees of freedom. It can then be shown that $E(\exp\{-(1/2)(\mathbf{d} - \mathbf{a})'\Sigma^{-1}(\mathbf{d} - \mathbf{a})\}) = 2^{-K}$, and the corresponding variance is $3^{-K/2} - 2^{-K}$. A quick calculation of the coefficient of variation shows that it increases exponentially with K . A characteristic of samples from such distributions is that many values are small

and one or two are huge. Thus, over the ensemble $\{\mathbf{Y}_{0:t}^{(l)} : l = 1, \dots, L\}$, where $K = n(t + 1)$, the normalized ISMC weights tend to concentrate on a few (or, just one) ensemble members. This phenomenon is known as *degeneracy*. As will be seen below, degeneracy is also an issue, even in low-dimensional problems where there are multiple time points.

8.4.5 Sequential Monte Carlo

In principle, a general sequential Monte Carlo (MC) algorithm follows from the usual sequential-update distributions, namely the forecast distribution (8.6) and the filtering distribution (8.7). As mentioned in Section 8.4.4, the choice of the prior distribution as the importance distribution gives weights proportional to the likelihood. We note in this case that these weights can be computed iteratively, since from (8.2) we obtain

$$v_t^{(l)} = p(\mathbf{Z}_{1:t} | \mathbf{Y}_{0:t}^{(l)}) = \prod_{k=1}^t p(\mathbf{Z}_k | \mathbf{Y}_k^{(l)}) = p(\mathbf{Z}_t | \mathbf{Y}_t^{(l)}) v_{t-1}^{(l)}. \quad (8.68)$$

This allows the importance sampler to be implemented sequentially, as we now discuss.

Assume we have L samples (or “particles”) from the importance distribution, $q(\cdot | \mathbf{Z}_{1:t})$, which here is the prior $p(\cdot)$, but need not be in general. Associated with each particle, $\mathbf{Y}_{0:t-1}^{(l)}$, is a normalized weight $w_{t-1}^{(l)}$; $l = 1, \dots, L$. Then from (8.6), the forecast density can be estimated as

$$\hat{p}(\mathbf{y}_t | \mathbf{Z}_{1:t-1}) \equiv \sum_{l=1}^L w_{t-1}^{(l)} p(\mathbf{y}_t | \mathbf{Y}_{t-1}^{(l)}),$$

where recall that $\mathbf{Y}_{0:t-1}^{(l)} = \{\mathbf{Y}_0^{(l)}, \dots, \mathbf{Y}_{t-1}^{(l)}\}$, and the conditional density, $p(\mathbf{y}_t | \mathbf{Y}_{t-1}^{(l)})$, on the right-hand side is known from the evolution model. Then, from (8.7) and (8.68), the filtering distribution at time t can be estimated as follows:

$$\hat{p}(\mathbf{y}_t | \mathbf{Z}_{1:t}) \propto p(\mathbf{Z}_t | \mathbf{y}_t) \sum_{l=1}^L v_{t-1}^{(l)} p(\mathbf{y}_t | \mathbf{Y}_{t-1}^{(l)}) = \sum_{l=1}^L v_t^{(l)} p(\mathbf{y}_t | \mathbf{Y}_{t-1}^{(l)}),$$

where, without loss of generality, the unnormalized weights are used.

Unfortunately, the importance weights will eventually degenerate as t increases (i.e., only a few samples, or one sample, will get all of the weight) and the posterior distribution will not be adequately represented by the sample. Various practical solutions to this problem have been considered; the primary one considered here is *particle filtering*.

8.4.6 Particle Filtering

In the presentation given in the previous subsections, the “particles” have been deemphasized, in favor of obtaining kernel estimates of the desired posterior distributions. In this subsection, the L particles, $\{\mathbf{Y}_{0:t}^{(1)}, \dots, \mathbf{Y}_{0:t}^{(L)}\}$, which are generated from the posterior distribution, $p(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t})$, are featured. It is conventional in the particle-filtering literature to represent the “density” by $\hat{p}(\mathbf{y}_{0:t}|\mathbf{Z}_{1:t}) = (1/L) \sum_{l=1}^L \delta_{\mathbf{Y}_{0:t}^{(l)}}$, where $\delta_{\mathbf{y}}$ is the Dirac delta function centered at $\mathbf{y}$. A comprehensive overview of the particle-filtering literature can be found in Doucet, de Freitas, and Gordon (2001). Overcoming the degeneracy-of-weights problem mentioned in Section 8.4.5 is crucial for practical implementation of such approaches.

There are many other approaches to dealing with the particle-filter degeneracy problem, and it remains an active area of research. Perhaps the most basic of these algorithms is the sampling importance resampling (SIR) or *bootstrap filter* (Doucet, de Freitas, and Gordon, 2001; Gordon, Salmond, and Smith, 1993). This algorithm accentuates the importance of particles with large weights and neglects the particles with small weights at each time step. The basic *SIR algorithm* can be written in pseudo code as follows:

1. Initialization at $t = 0$
 - For $l = 1, \dots, L$, sample $\mathbf{Y}_{0|0}^{(l)} \sim p(\mathbf{y}_0)$ and set $t = 1$.
2. Importance-sampling step
 - For $l = 1, \dots, L$, sample $\tilde{\mathbf{Y}}_t^{(l)} \sim p(\mathbf{y}_t|\mathbf{Y}_{t-1}^{(l)})$, and set $\tilde{\mathbf{Y}}_{0:t}^{(l)} = \{\mathbf{Y}_{1:t-1}^{(l)}, \tilde{\mathbf{Y}}_t^{(l)}\}$.
 - For $l = 1, \dots, L$, evaluate the importance weights, $\tilde{w}_t^{(l)} = p(\mathbf{Z}_t|\tilde{\mathbf{Y}}_t^{(l)})$. (Note that in this algorithm these weights are *not* proportional to the weights at the previous time, $t - 1$, due to the resampling in step 3 below, which induces equal weights on the resample.)
 - Normalize the IS weights.
3. Resampling step
 - Resample, with replacement, L particles $\{\mathbf{Y}_{0:t}^{(l)} : l = 1, \dots, L\}$ from the set $\{\tilde{\mathbf{Y}}_{0:t}^{(l)} : l = 1, \dots, L\}$ using the importance weights for the resampling.
 - Set $t = t + 1$ and go to step 2.

The resampling step in the SIR algorithm reduces the effects of sample degeneracy. However, since particles with large weights are selected many times, there can be a loss of diversity among the particles, leading to “sample impoverishment” (Arulampalam et al., 2002). Pitt and Shepard (1999) considered a variant of the basic bootstrap filter, which they called “auxiliary sampling importance resampling” (ASIR), to address this problem. The basic idea is that one seeks to augment “good” particles such that the associated

predictive likelihoods are large for these particles. Thus, the ASIR filter reduces the cost of sampling by resampling less frequently, particles that have small likelihoods. In the actual ASIR implementation, an auxiliary variable is introduced that allows the filter to adapt to the data more efficiently. Such filters, although an improvement, still suffer degeneracy in high-dimensional problems and when the process noise is relatively large. The incorporation of Markov chain moves into the importance sampling algorithm is an alternative approach to improving the degeneracy issues in traditional particle filters. The primary goal is to reduce the variability of the importance weights by attempting more accurate moves at each time step (e.g., Berzuini et al., 1997; MacEachern, Clyde, and Liu, 1999; Godsill and Clapp, 2001; Polson, Stroud, and Müller, 2008).

The extension of the particle filter to parameter estimation is nontrivial. One approach is to model the parameters as a random walk and then augment the state space with the parameters. A limitation with this approach is that it assumes that the state and the parameters have the same conditional-distributional form (e.g., Gaussian), where in reality certain parameters may be constrained in ways to suggest different distributions are more appropriate. Another limitation of this method is that it implies an increase in the variance of the parameters with time, resulting in posteriors that are more diffuse than the actual ones; shrinkage kernels can be used to address this, where the variance of parameters is based on particles from previous time points (Liu and West, 2001). Berliner and Wikle (2007) consider a number of approaches that can help when performing parameter and state process estimation with importance sampling. These include dimension reduction approaches that project processes and parameters into lower dimensional spaces, as well as the use of sufficient statistics and various approximations. Carvalho et al. (2010) describe a promising approach that also exploits the use of sufficient statistics. One can also use Markov chain Monte Carlo moves to facilitate parameter estimation (e.g., Polson, Stroud, and Müller, 2008). In particular, the hybrid sequential Monte Carlo and MCMC approaches of Andrieu, Doucet, and Holenstein (2010) appear to allow construction of efficient proposal distributions for high-dimensional, highly correlated processes using sequential importance sampling within the MCMC algorithm, in a manner that ensures convergence.

8.4.7 Ensemble Kalman Filtering

Although the algorithms and approaches described above help with the degeneracy issue, experience has shown that in high-dimensional situations, like one might encounter in real-world spatio-temporal problems, degeneracy is still problematic for sequential-importance-sampling methodologies. A closely related idea is to use MC in the context of the Kalman filter. Such methods are typically referred to as *ensemble Kalman filters*.

The ensemble Kalman filter (EnKF) was originally developed by Evensen (1994), Evensen and van Leeuwen (1996), and Houtekamer and Mitchell

(1998). The basic approach uses Monte Carlo samples from Gaussian distributions to *approximate* the forecast distribution, while, critically, still using the nonlinear process-evolution model. In particular, one estimates the prior (forecast) means and covariance matrices with the Monte Carlo sample (ensemble). This is then used in the linear KF update formulas to obtain the filter distribution. In general, the EnKF algorithm is like the particle filter, with a set of particles being propagated forward with the dynamic model. However, instead of computing weights and resampling, the state-process vectors are updated using the Kalman gain matrix (as in traditional Kalman filtering), but with the forecast sample mean and sample covariance matrix from the ensemble of particles. Importantly, calculation of the sample covariance matrix must be “regularized” (in some way) to accommodate the relatively small number of particles used in its estimation. Although EnKF algorithms are typically stable for high-dimensional state variables, and they do not have the same sample impoverishment problems as the traditional particle filter, they are approximations and still often require parameter-estimation procedures that are disjoint from state estimation. Stroud et al. (2010) provide a discussion of such methods applied to sediment transport modeling in Lake Michigan using an advection-diffusion model (which fits into the nonlinear GQN framework discussed in Section 7.3.3) to motivate the latent dynamical process. In addition, more details can be found in the overview presented in Wikle and Berliner (2007) and Berliner and Wikle (2007), where an example of the EnKF on a simulated AR(1) dataset (similar to that presented in Section 8.2.1) can be found.

Berliner and Wikle (2007) also discuss many other approximate importance sampling algorithms, particularly those focused on exploiting reduced dimensionality.

8.4.8 Integrated Nested Laplace Approximation (INLA)

Finally, a method for implementing latent Gaussian models that has potential in the DSTM context is based on the integrated nested Laplace approximation (i.e., INLA; Rue and Held, 2005; Rue, Martino, and Chopin, 2009; Ruiz-Cárdenas, Krainski, and Rue, 2010). The INLA approach is a numerically implemented analytical solution for approximating posterior marginals in hierarchical models with latent Gaussian processes. The advantages of INLA over Monte Carlo-based solutions (e.g., importance sampling, MCMC) are primarily that INLA algorithms can be orders of magnitude faster and automatic, thus reducing the need for algorithm tuning and making it easier (and quicker) to build and check models. In practice, INLA methods appear to be accurate and, due to the availability of INLA software (e.g., Martino and Rue, 2009), a wide selection of models that meet the INLA criteria can be easily fitted using a suite of previously developed “black box” codings. For models that have a large number of non-Gaussian hyperparameters, process nonlinearities, or spatial structures that cannot be coerced into a GMRF, it is not yet clear

if INLA methods will be directly helpful. However, new extensions are being rapidly developed and incorporated into existing INLA software.

We are hopeful that, in the near future, a combination of analytical/computational and Monte Carlo procedures will be available for implementing DSTMs within a hierarchical statistical model, where the models are more and more nonlinear and increasingly non-Gaussian.

8.5 BIBLIOGRAPHIC NOTES

Implementation and inference for DSTMs is rapidly evolving, given its reliance on MCMC and importance sampling algorithms. We fully realize that almost anything said here will be out of date by the time the book is published. However, there are some fundamental implementation details that have stood the test of time, and we have referred to many of them in this chapter. Because of space limitations, not a lot of detail can be given; for that, we refer the reader to many of the excellent books that deal with modern implementations of dynamical models, as well as more general sequential and MCMC implementations. Specifically, we recommend Shumway and Stoffer (2006) for an overview of state-space implementations of various types. We also note that Barndorff-Nielsen, Cox, and Klüppelberg (2001), Robert and Casella (2004) and Robert and Casella (2010) give comprehensive overviews of general Monte Carlo approaches, and Doucet, de Freitas, and Gordon (2001) give an overview of practical sequential Monte Carlo approaches. West and Harrison (1997), Gamerman and Lopes (2006), Marin and Robert (2007), Kitagawa (2010) and Prado and West (2010) give modern perspectives on Bayesian implementation of dynamical models. Banerjee, Carlin, and Gelfand (2004), Clark and Gelfand (2006), Le and Zidek (2006), Diggle and Ribeiro (2007), and Finkenstädt, Held, and Isham (2007) give specific implementational details for various spatial and spatio-temporal models. In addition, we recommend Clark (2007) as an excellent introduction to various inference approaches to modeling complicated data and processes. Finally, we highly recommend the book by Royle and Dorazio (2008), which contains a similar hierarchical modeling approach as that presented in our book, and it gives a variety of detailed examples at various levels, along with specific implementation details.

Sequential Process Estimation

Many of the references for statistical time series analysis given in Chapter 3 discuss sequential estimation to various degrees. Our presentation follows that given in Shumway and Stoffer (2000, 2006). However, the seminal works of Kalman (1960), Jazwinski (1970), Anderson and Moore (1979), West and Harrison (1997), and Doucet, de Freitas, and Gordon (2001) are also excellent references. For the Bayesian perspective, we rely on Jazwinski (1970), Meinhold and Singpurwalla (1983), West and Harrison (1997), and Doucet, de Freitas, and Gordon (2001). We have also found the books by Grewal and

Andrews (1993), Eubank (2006), Kitagawa (2010) and Prado and West (2010) particularly helpful.

Parameter Estimation in Sequential Implementations

Parameter estimation in sequential implementations is covered in many classic texts, but in this chapter we focus on Gupta and Mehra (1974), Carter and Kohn (1994), Frühwirth-Schnatter (1994), West and Harrison (1997), Shumway and Stoffer (1982, 2000, 2006), Doucet, de Freitas, and Gordon (2001), Robert and Casella (2004), Xu and Wikle (2007), and Katzfuss and Cressie (2011).

Particle Filtering

Particle filtering (and its various derivatives) is a rapidly developing area of research. We have presented just the basics in this chapter, motivated by Gordon, Salmond, and Smith (1993), Pitt and Shephard (1999), and Doucet, de Freitas, and Gordon (2001). Irwin, Cressie, and Johannesson (2002) give a comparative review of the Unscented Particle Filter and the Extended Kalman Filter. We also discussed, briefly, the ensemble Kalman filter, which is an approximate sequential importance sampling approach developed in the atmospheric and oceanographic sciences. This is also a rapidly developing area, and we have cited a few of the fundamental references in the area: Evensen (1994), Evensen and van Leeuwen (1996), Houtekamer and Mitchell (1998), Evensen (2009), and Stroud et al. (2010). We have also borrowed notation and examples from the tutorial paper by Wikle and Berliner (2007).

Combined MCMC/ISMC

The combined use of MCMC and ISMC may be very useful in nonlinear and non-Gaussian DSTMs, as suggested by the work of Andrieu, Doucet, and Holenstein (2010). Historically, there have been several approaches to combining MCMC and ISMC. Berzuini et al. (1997) consider Markov chain Monte Carlo (MCMC) moves within the particle filter, which they call the Metropolis–Hastings Importance Resampling (MHIR) method. MacEachern, Clyde, and Liu (1999) incorporated Markov chain transitions into a sequential importance sampling algorithm for a nonparametric Bayes model. Berliner, Milliff, and Wikle (2003) proposed a hybrid MCMC/ISMC algorithm for a multivariate spatio-temporal process for the ocean/atmosphere system. In this case, MCMC samples were used from the atmospheric portion of the model within the ISMC ocean model, with a final re-weighting of the atmospheric samples based on the importance weights from the ocean model. Godsill and Clapp (2001) proposed a method to incorporate MCMC resampling ideas into the particle filter by making the MCMC traverse a sequence of “bridging densities” that lie between the forecast density [see (8.4)] and the filtering density [see (8.5)]. Polson, Stroud, and Müller (2008) proposed an approach to Bayesian filtering and sequential parameter learning in general state-space models that relies on a rolling-window MCMC algorithm; the algorithm approximates the target posterior distribution by a mixture of lag- k smoothing distributions.

MCMC Implementations of Nonlinear DSTMs

It is worth noting that one can develop more efficient “block” updates of the state process and the parameters for Metropolis–Hastings implementations of nonlinear (nonconjugate) models. A very useful example is the so-called “differential evolution MCMC” approach of ter Braak (2006), the “delayed rejection adaptive Metropolis” approach of Haario et al. (2006), and the “differential evolution adaptive Metropolis” algorithm of Vrugt et al. (2009). These algorithms and others are evaluated in Higdon et al. (2008).

Alternative Implementations

In terms of other useful methods for fitting nonlinear models, such as the GQN models in Section 7.3.3, when only characteristics of posterior distributions are of interest, specific numerical-optimization procedures have been developed to estimate posterior modes in the nonlinear state-space-modeling framework (So, 2003). Another approach that may prove to be useful is referred to as variational Bayesian learning (e.g., Raiko et al., 2007). These methods, which are largely dependent on additive Gaussian error in both the data model and the process model, can be employed in situations where there exist significant nonlinearities in the mean of the HM components. In essence, these methods work by approximating the intractable posterior distribution of interest with a tractable parametric distribution, and then the estimation is corrected through the use of a cost function that is based on the discrepancy or misfit between the distributions (often Kullback–Leibler divergence). Such methods have proven valuable in applications involving Control Theory (e.g., Raiko and Tornio, 2009) and, as they continue to develop, they may prove to be useful for general DSTMs. In addition, Laplace approximations can provide fast approximate state process estimates in some nonlinear and non-Gaussian settings (Koyama et al. 2010).